



Alliance for Biking & Walking

BICYCLING AND WALKING IN THE UNITED STATES

2012

BENCHMARKING REPORT





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PREFACE

Alliance for Biking & Walking

Alliance for Biking & Walking is the North American coalition of grassroots bicycling and walking advocacy organizations. Our mission is to create, strengthen, and unite state/provincial and local bicycle and pedestrian advocacy organizations. Since our founding in 1996, we have grown from 12 to nearly 200 member organizations representing 48 U.S. states, four Canadian provinces, and Mexico City.

In the last 16 years, we have improved the effectiveness of our organizations through trainings and the sharing of best practice models in organizational development and bicycling and walking initiatives. We are continually improving our delivery channels through executive coaching, replicable models, trainings, our on-call support system, strategic planning, and resources like this report.

Alliance organizations inform and organize their communities to improve conditions for bicycling and walking, promoting these as healthy and enjoyable ways to travel. From advocating for bikeways and walkways to conducting safety courses, our coalition is changing attitudes and the environment in communities across North America. The Alliance connects these grassroots forces by fostering peer networking and supporting each other in our efforts to promote bicycling and walking for healthy communities, a healthy environment, and a better quality of life.

Benchmarking Project Origins

The Alliance's Benchmarking Project began in 2003 when Alliance leaders recognized the need for advocates to measure progress of bicycling and walking and realized the lack of available data. Our staff and board jumped



on the project, recognizing the benefit of showing the impact advocacy has on increasing bicycling and walking. Without hard data to measure results, Alliance organizations were missing a key argument for their efforts.

The Benchmarking Project was also aligned with public health organizations and objectives. The process of benchmarking is designed to facilitate communities to build healthy and safe community environments. This is one of four key directions outlined by the U.S. Department of Health and Human

The process of
benchmarking is
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safe community
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Services as fundamental to bring prevention into our communities. Furthermore, it aligns with Centers for Disease Control and Prevention's (CDC) Winnable Battles to reduce motor vehicle injuries and increase physical activity. The partnerships addressed in this report among bicycle and pedestrian groups, health organizations, and transportation are necessary to address the infrastructure problems in our communities to improve public health, in the same way that municipal water systems and improved housing infrastructure helped

remove infectious disease risks in the previous century.

In 2004 the Alliance completed a pilot benchmarking report collecting data only on bicycling from just 15 cities and 15 states to test methods for the project. This first report helped pave a smoother path for the collection of more comprehensive data from all 50 states and 50 cities in 2006 and 2007. The first full report on the status of bicycling and walking in the United States was published in August 2007 (under the organization's former name: Thunderhead Alliance). The second full report was published in January 2010. This second report marked the first time we had in place a system to track usage and dissemination of report findings. To date, nearly 6,000 copies of the report have been downloaded or distributed. The 2010 report was cited or referenced in over 300 media stories, reports, plans, and articles.

This third full report builds upon our previous efforts to deliver timely data to help locals measure their progress and effectiveness, set new goals, and achieve greater results.

Through the ongoing Benchmarking Project, the Alliance for Biking & Walking will publish an updated version of this report every two years and will continuously refine methods and consider new data sets as available. As the project progresses, it will offer more precise benchmarks and recommendations for advocates and government officials so that they have the data they need to improve bicycling and walking in the United States and eventually all of North America.



Photo by Teena Widman

EXECUTIVE SUMMARY

What isn't counted, doesn't count.

Government officials working to promote bicycling and walking need data to evaluate their efforts. In order to improve something, there must be a means to measure it. The Alliance for Biking & Walking's Benchmarking Project is an ongoing effort to collect and analyze data on bicycling and walking in all 50 states and the 51 largest cities. This is the third biennial Benchmarking Report. The first report was published in 2007, the second in 2010, and the next report is scheduled for January 2014.

Objectives

(1) Promote Data Collection and Availability

The Benchmarking Project aims to collect data from secondary sources (existing databases) and to conduct surveys of city and state officials to obtain data not collected by another national source. A number of government and national data sources are collected and illustrated in this report. Through state and city biennial surveys, this project makes new data available in a standardized format that otherwise does not exist.

(2) Measure Progress and Evaluate Results

The Benchmarking Project aims to provide data to government officials and advocates in an accessible format that helps them measure their progress toward increasing bicycling and walking and evaluate the results of their efforts. Because the Benchmarking Project is ongoing, cities and states can measure their progress over time and will see the impacts of their efforts. By providing a consistent and objective tool for evaluation, organizations, states, and cities can determine what works and what doesn't. Successful models can be emulated and failed models reevaluated.

(3) Support Efforts to Increase Bicycling and Walking

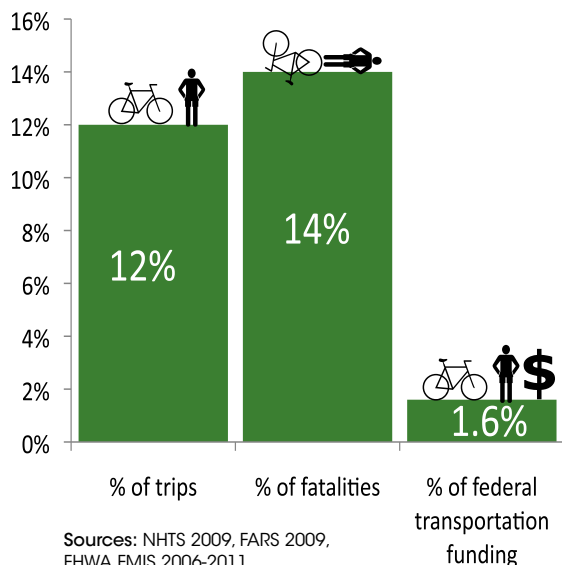
This project will ultimately support the efforts of government officials and bicycle and pedestrian advocacy organizations to increase bicycling and walking in their communities. By providing a means for cities and states

to compare themselves to one another, this report will highlight successes, encourage communities making progress, and make communities aware of areas where more effort is needed. By highlighting the top states and cities, other states and cities will gain inspiration and best practice models. This report is intended to help states and communities set goals, plan strategies, and evaluate results.

Data Collection

This report focuses on 50 states and the 51 largest U.S. cities. Most bicycling and walking is in urban areas, and because of short trip distances, the most potential for increasing bicycling and walking is in cities. Whenever possible, the Alliance collected data for this report directly from uniform government data sources. Researchers collected data that were not readily accessible from national sources through two surveys for cities and states. In October 2010,

Levels of Bicycling and Walking,
Bike/Ped Fatalities, and
Bike/Ped Funding in the U.S.



Bicyclists and pedestrians make up 12% of all trips and account for 14% of traffic fatalities, yet just 1.6% of federal transportation funds go to these modes.

Overview of Walking, Bicycling, Transit, and Car Mode Share

Mode of Travel	% of Commuters ⁽¹⁾		% of All Trips Nation-wide ⁽²⁾
	Nation-wide	Major U.S. Cities	
	2.9%	4.9%	10.5%
	0.6%	0.9%	1.0%
	5.0%	17.2%	1.9%
	91.5%	77.0%	86.6%

Sources: (1) ACS 2009 (2) NHTS 2009 Notes: (3) This includes trips by private car and "other" means that are not public transportation, bicycling, or walking.

the Benchmarking Project team reached out to 50 states and 51 cities, utilizing the staff of cities, state departments of transportation, metropolitan planning organizations, and advocacy organizations to provide data for city and state surveys. The surveys complemented existing government data sources to create a comprehensive reserve of data that evaluates multiple factors that affect bicycling and walking in cities and states.

Results

Levels of Bicycling and Walking

From 1990 to 2009, the percent of commuters who bicycle to work increased from 0.4% to 0.6% while the percent of commuters who walk to work decreased from 3.9% to 2.9%. According to the 2009 American Community Survey (ACS), 3.4% of commuters nationwide are bicyclists (0.55%) or pedestrians

Changes 2005-2010

	05/06	07/08	09/10
KEY FIGURES			
Percent of commuters who walk	2.5%	2.8%	2.9%
Percent of commuters who bicycle	0.4%	0.5%	0.6%
Percent of commuters who walk or bicycle	2.9%	3.3%	3.4%
Percent of traffic fatalities: ped	11.2%	11.3%	11.7%
Percent of traffic fatalities: bicycle	1.7%	1.8%	1.8%
Number of ped traffic fatalities	4,892	4,699	4,092
Number of bicycle traffic fatalities	786	701	630
Percent of fed. trans. \$ to bike/ped	1.5%	1.2%	1.6%
Number of states/cities responding ⁽²⁾	46/45	47/48	48/48
STATE POLICIES (Number of states with)			
Goal to increase walking	16 ⁽¹⁾	22	35
Goal to increase bicycling	16 ⁽¹⁾	21	35
Goal to decrease ped fatalities	18 ⁽¹⁾	31	41
Goal to decrease bicycle fatalities	18 ⁽¹⁾	31	38
Bicycle advisory committee	*	20	24
Pedestrian advisory committee	*	18	22
Bicycle master plan	*	27	27
Pedestrian master plan	*	24	25
Safe passing legislation	*	14	21
Complete streets policy	10	17	24
CITY POLICIES (Number of cities with)			
Goal to increase walking	25 ⁽¹⁾	20	36
Goal to increase bicycling	25 ⁽¹⁾	33	46
Goal to decrease ped fatalities	20 ⁽¹⁾	19	31
Goal to decrease bicycle fatalities	20 ⁽¹⁾	26	39
Bicycle advisory committee	*	33	36
Pedestrian advisory committee	*	31	26
Bicycle master plan	*	36	42
Pedestrian master plan	*	12	22
Complete streets policy	8	13	18
STATE PROVISIONS			
Per capita \$ to bike/ped	\$2.50	\$1.29	\$2.17
CITY PROVISIONS			
Per capita \$ to bike/ped	\$1.83	\$1.49	\$1.80
Miles bicycle facilities/sq. mile	1.3	1.4	1.8
Bike parking at transit/10K people	1.7	2.5	2.5
% buses with bike racks	69%	93%	95%
STATE EDUCATION & ENCOURAGEMENT (Number of states with)			
Annual state bike/ped conference	*	15	25
Drivers test questions on bicycling	*	23	32
Share the road/safety campaign	*	33	38
CITY EDUCATION & ENCOURAGEMENT (Number of cities with)			
Youth bike ed courses	*	29	38
Adult bike ed courses	*	33	41
Bike to Work Day events	*	37	43
Open street (ciclovía) initiatives	*	12	21
City-sponsored bike ride	*	23	32
OTHER			
States with dedicated advocacy org	32	35	43
Cities with dedicated advocacy org	32	34	36

(1) Walking and bicycling were combined in this survey (2) Number of states/cities who responded to the Benchmarking Report survey
* = Data unavailable

High to Low Ranking of Bicycling and Walking Levels

STATES

1. Alaska
2. Vermont
3. New York
4. Montana
5. Oregon
6. Hawaii
7. Massachusetts
8. South Dakota
9. Wyoming
10. Maine
11. North Dakota
12. Pennsylvania
13. Idaho
14. Iowa
15. Washington
16. Colorado
17. Wisconsin
18. Minnesota
19. California
20. Illinois
21. Nebraska
22. Utah
23. New Jersey
24. New Hampshire
25. Rhode Island
26. Connecticut
27. Kansas
28. New Mexico
29. Arizona
30. West Virginia
31. Nevada
32. Maryland
33. Delaware
34. Michigan
35. Indiana
36. Virginia
37. Kentucky
38. Ohio
39. Louisiana
40. Missouri
41. Oklahoma
42. Florida
43. North Carolina
44. South Carolina
45. Texas
46. Mississippi
47. Arkansas
48. Georgia
49. Tennessee
50. Alabama

CITIES

1. Boston
2. Washington, DC
3. San Francisco
4. Seattle
5. New York
6. Portland, OR
7. Minneapolis
8. Philadelphia
9. Honolulu
10. New Orleans
11. Baltimore
12. Chicago
13. Oakland
14. Denver
15. Sacramento
16. Tucson
17. Milwaukee
18. Atlanta
19. Cleveland
20. Los Angeles
21. Miami
22. Long Beach
23. San Diego
24. Detroit
25. Columbus
26. Albuquerque
27. Austin
28. Raleigh
29. Colorado Springs
30. Mesa
31. Omaha
32. San Jose
33. Louisville
34. Fresno
35. Virginia Beach
36. Tulsa
37. Houston
38. Phoenix
39. Indianapolis
40. Las Vegas
41. Kansas City, MO
42. El Paso
43. Memphis
44. Charlotte
45. San Antonio
46. Arlington, TX
47. Nashville
48. Jacksonville
49. Dallas
50. Oklahoma City
51. Fort Worth

(2.86%). Residents of major U.S. cities are 1.7 times more likely to walk or bicycle to work than the national average. According to the 2009 National Household Travel Survey (NHTS) 1.0% of all trips are by bicycle and 10.5% of all trips are by foot nationwide. It is difficult to determine bicycling and walking mode share for all trips at the state and city levels because of small sample sizes of NHTS.

Bicycle and pedestrian commuters are generally distributed proportionately among ethnic groups in the U.S., according to the 2009 ACS. Greater disparities are found among genders. According to the 2009 NHTS, 49% of walking trips are men and 51% are female, yet among bicycle trips, 76% are male and only 24% are female. A look at age reveals that while walking is generally distributed proportionately among age groups, youth under age 16 make up 39% of bicycle trips. This age group accounts for just 21% of the population.

Safety

In 2009, 4,092 pedestrians and 630 bicyclists were killed in traffic. This is down significantly from 2005 when 4,892 pedestrians and 786 bicyclists were traffic fatality victims. While overall numbers of bicycle and pedestrian fatalities are declining, pedestrians and bicyclists are still at a disproportionate risk for being a victim of a traffic fatality. Although just 10.5% of trips in the U.S. are by foot and 1.0% are by bicycle, 11.7% of traffic fatalities are pedestrians and 1.8% are bicyclists. In major U.S. cities, 12.7% of trips are by foot and 1.1% are by bicycle,

Source: 2007-2009 ACS Notes: This ranking is based on the combined bike and walk to work share from the 2007-2009 ACS. The number one position indicates the state and city with the highest share of commuters who commute by bicycle or foot. View graphs illustrating this data on pages 34 and 35 of this report.

yet 26.9% of traffic fatalities are pedestrians and 3.1% are bicyclists.

According to the 2007-2009 Fatality Analysis Reporting System (FARS) and the 2009 NHTS, seniors are the most vulnerable age group. While adults over 65 make up 10% of walking trips and 6% of bicycling trips, they account for 19% of pedestrian fatalities and 10% of bicyclist fatalities.

Policies and Provisions

A number of policies and provisions are represented in this report including funding and staffing levels, infrastructure, written policies, and bike-transit integration. This report marks a significant increase in planning for bicycling and walking over the last two years. Many states and cities have adopted new plans and goals to increase bicycling and walking and reduce fatalities. Overall, states and cities still rank poorly for funding bicycling and walking at a rate proportionate to active transportation levels.

Funding for Bicycling and Walking

2010 data from the Federal Highway Administration reveal that states spend just 1.6% of their federal transportation dollars on bicycling and walking. This amounts to just \$2.17 per capita for bicycling and walking. About 40% of these dedicated bicycle and pedestrian dollars are from the Transportation Enhancement (TE) program. The majority of TE funding (48%) goes toward building bicycle and pedestrian facilities and to bicycle and pedestrian education.

Sources: FARS 2007-2009 ACS 2007-2009 **Notes:** This ranking is based on the fatality rate which is calculated by dividing the number of annual pedestrian and bicycle fatalities (averaged between 2007-2009) by population (weighted, or multiplied, by share of the population walking and bicycling to work). The number one position indicates the safest state or city according to the fatality rate. View these data on pages 56-62 of this report.

Low to High Ranking of Bike/Ped Fatality Rates

STATES

1. Vermont
2. Nebraska
3. Alaska
4. Wyoming
5. South Dakota
6. North Dakota
7. Iowa
8. Maine
9. Massachusetts
10. Minnesota

11. Idaho
12. New Hampshire
13. Oregon
14. Washington
15. Wisconsin
16. Montana
17. New York
18. Pennsylvania
19. Kansas
20. Colorado
21. Hawaii
22. Utah
23. Illinois
24. Connecticut
25. West Virginia
26. Ohio
27. Indiana
28. Rhode Island
29. Virginia
30. Kentucky
31. New Jersey
32. California
33. Michigan
34. Missouri
35. Oklahoma
36. Nevada
37. Tennessee
38. New Mexico
39. Maryland
40. Arizona
41. Arkansas
42. Delaware
43. Texas
44. North Carolina
45. Georgia
46. Mississippi
47. Alabama
48. Louisiana
49. South Carolina
50. Florida

CITIES

1. Boston
2. Minneapolis
3. Omaha
4. Seattle
5. Portland, OR
6. Washington, DC
7. New York
8. San Francisco
9. Philadelphia
10. Honolulu

11. Colorado Springs
12. Chicago
13. Cleveland
14. Oakland
15. Baltimore
16. Milwaukee
17. Sacramento
18. Denver
19. Virginia Beach
20. Tucson
21. Mesa
22. San Diego
23. New Orleans
24. San Jose
25. Columbus
26. Los Angeles
27. Atlanta
28. Indianapolis
29. Long Beach
30. Austin
31. Arlington, TX
32. Raleigh
33. Albuquerque
34. Las Vegas
35. El Paso
36. Memphis
37. Fresno
38. San Antonio
39. Nashville
40. Detroit
41. Houston
42. Charlotte
43. Louisville
44. Miami
45. Kansas City, MO
46. Oklahoma City
47. Tulsa
48. Phoenix
49. Dallas
50. Jacksonville
51. Fort Worth

High to Low Ranking of Per Capita Funding to Bike/Ped

STATES

1. Alaska
2. Vermont
3. Montana
4. Wyoming
5. Delaware
6. New Mexico
7. South Dakota
8. Rhode Island
9. Kentucky
10. Iowa
11. Minnesota
12. North Dakota
13. Indiana
14. Washington
15. Pennsylvania
16. New Hampshire
17. Missouri
18. Tennessee
19. Idaho
20. Maine
21. Florida
22. Arizona
23. Georgia
24. Oregon
25. Massachusetts
26. Alabama
27. North Carolina
28. Louisiana
29. Kansas
30. Colorado
31. Utah
32. Hawaii
33. Michigan
34. California
35. Connecticut
36. Nebraska
37. New York
38. Mississippi
39. West Virginia
40. Texas
41. Ohio
42. Wisconsin
43. Arkansas
44. Nevada
45. Illinois
46. South Carolina
47. New Jersey
48. Oklahoma
49. Virginia
50. Maryland

CITIES

1. Washington, DC
2. Minneapolis
3. Sacramento
4. Miami
5. Tucson
6. Dallas
7. New Orleans
8. Albuquerque
9. Nashville
10. Oakland
11. Atlanta
12. Kansas City, MO
13. Portland, OR
14. Omaha
15. San Diego
16. Philadelphia
17. Raleigh
18. San Francisco
19. Indianapolis
20. Houston
21. San Jose
22. San Antonio
23. Charlotte
24. Denver
25. Austin
26. Milwaukee
27. Jacksonville
28. Memphis
29. Long Beach
30. Phoenix
31. Fresno
32. El Paso
33. Seattle
34. Detroit
35. Colorado Springs
36. Louisville
37. Columbus
38. Cleveland
39. Tulsa
40. Honolulu
41. Los Angeles
42. Boston
43. Fort Worth
44. Mesa
45. Baltimore
46. Chicago
47. Virginia Beach
48. Arlington, TX
49. Las Vegas
50. New York

Planning and Legislation

Since the 2010 Benchmarking Report, there has been a 63% increase in the number of states that have published goals to increase bicycling and walking, and a 27% increase in the number of states that have published goals to reduce bicycle and pedestrian fatalities.

2011 League of American Bicyclist data on state legislation reveal that most states have basic bicyclists' rights legislation such as allowing bicyclists to legally ride two-abreast, signal right turns with their right hand, and to take a full traffic lane in the presence of a sidepath or bike lane. Twenty-one states have 3-foot passing laws that require motorists to pass bicyclists at a safe distance of at least three feet (up from 14 as of the 2010 Benchmarking Report).

A survey of other policies found that 19 (of the 51 largest) U.S. cities and 26 states have adopted complete streets policies that require streets be built to accommodate all potential road users. Nearly half of states report having a bicycle and pedestrian advisory committee. And 38 states report having a publicly available bicycle map.

Cities were surveyed on a number of planning and policy initiatives. Forty-one cities report having a bicycle master plan, and 21 have a pedestrian master plan. Over half of cities have bicycle and pedestrian advisory committees. (Continued page 16)

Source: FHWA FMIS 2006-2010 **Notes:** This ranking is based on the per capita spending of federal funds by states and cities on bicycling and walking using a 5-year average (2006-2010). Data is based on funds obligated to projects in this period and are not necessarily the amount spent in these years. The number one position indicates the state or city with the highest amount of per capita federal funding to bicycling and walking. Due to large amounts of deobligated funds in the 5-year period between 2006-2010, accurate funding estimates could not be obtained for Oklahoma City. View these data on pages 86-87 of this report.

State Overview of Primary Benchmarking Indicators

Key: ● = Top 1/3 among states ○ = Middle 1/3 among states ○ = Bottom 1/3 among states * = data unavailable

State	Mode Share	Safety	Funding	Policy (1)	Education/Encouragement (2)	Advocacy Capacity (3)
Alabama	○	○	●	○	○	●
Alaska	●	●	●	●	○	○
Arizona	●	○	●	●	●	●
Arkansas	○	○	○	○	○	●
California	●	●	●	●	●	●
Colorado	●	●	●	●	●	●
Connecticut	●	●	○	●	●	●
Delaware	●	○	●	●	●	●
Florida	○	○	●	●	●	●
Georgia	○	○	●	●	○	●
Hawaii	●	●	●	●	○	●
Idaho	●	●	●	●	○	●
Illinois	●	●	○	●	●	●
Indiana	○	●	●	○	●	*
Iowa	●	●	●	○	●	●
Kansas	●	●	●	●	●	*
Kentucky	○	●	●	●	●	○
Louisiana	○	○	●	●	●	○
Maine	●	●	●	●	●	●
Maryland	●	○	○	●	●	●
Massachusetts	●	●	●	●	●	●
Michigan	●	●	●	●	○	●
Minnesota	●	●	●	●	●	●
Mississippi	○	○	○	○	●	●
Missouri	○	●	●	○	●	●
Montana	●	●	●	*	*	○
Nebraska	●	●	○	○	○	○
Nevada	●	○	○	●	●	●
New Hampshire	●	●	●	○	●	●
New Jersey	●	●	○	●	●	●
New Mexico	●	○	●	*	*	●
New York	●	●	○	●	○	●
North Carolina	○	○	●	●	●	●
North Dakota	●	●	●	○	○	○
Ohio	○	●	○	○	●	*
Oklahoma	○	○	○	○	●	●
Oregon	●	●	●	●	●	●
Pennsylvania	●	●	●	●	○	●
Rhode Island	●	●	●	●	●	*
South Carolina	○	○	○	○	●	●
South Dakota	●	●	●	○	●	●
Tennessee	○	○	●	●	●	●
Texas	○	○	○	○	●	*
Utah	●	●	●	●	●	●
Vermont	●	●	●	●	●	*
Virginia	○	●	○	●	●	●
Washington	●	●	●	●	●	●
West Virginia	●	●	○	○	○	○
Wisconsin	●	●	○	●	●	●
Wyoming	●	●	●	○	●	○
Find the data (page)	45-46	56-57	86	68, 72, 80	111, 113	138

Interpreting the State and City Overview Tables

The tables on this page and next give an overview of how states and cities compare in six areas. Full circles indicate the best ranking; states and cities with full circles are within the top 1/3 among their peers. Half-circles represent the middle 1/3, and empty circles represent the bottom 1/3. States and cities with the most filled in circles represent those that are setting the benchmarks for bicycling and walking levels, safety, funding, policies, education/encouragement, and advocacy capacity. Below is an explanation for how the rankings on this page and next were determined.

Mode Share: This ranking is based on the combined share of commuters who bicycle and walk to work averaged over the most recent three years. The top 1/3 states and cities are those with the highest percentage of workers who commute by bicycle and foot. **Data source:** ACS 2007-2009.

Safety: This ranking is based on the bicycle and pedestrian fatality rate defined as number of annual bicycle and pedestrian deaths (using a 3-year average) divided by the population (weighted, or multiplied, by share of commuters who bicycle and walk to work). The top 1/3 states and cities are those with the lowest fatality rate, and thus the highest safety ranking. **Data Sources:** FARS 2007-2009, ACS 2009.

Funding: This ranking is based on the federal dollars per capita that are obligated to bicycling and walking annually. The top 1/3 states and cities are those with the highest per capita investment of federal dollars in bicycling and walking. **Data Source:** FHWA 2004-2008.

Policies: This ranking is based on the total number of policies promoting bicycling and walking adopted by the state/city. **Policies counted for states include:** Goals to increase walking, increase bicycling, decrease pedestrian fatalities, and decrease bicycle fatalities; Master Plan adopted for bicycling, walking, and trails; Bike/ Ped advisory committee; legal 2-abreast riding for bicycles; 3-foot/safe passing legislation; spending target; publicly available bicycle map; complete streets policy. **Policies counted for cities include:** goals to increase walking, increase bicycling, decrease pedestrian fatalities, and decrease bicycle fatalities; Master Plan adopted for bicycling and for walking; Bike/ Ped advisory committee; bicycle parking requirements in building/ garages, new buildings, and at public events; complete streets policy. **Data Sources:** State surveys, city surveys, League of American Bicyclists (1)

Education/Encouragement: This ranking is based on the total number of education/encouragement programs and state/city events. **Those counted for states include:**

City Overview of Primary Benchmarking Indicators

Key: ● = Top 1/3 among cities ◐ = Middle 1/3 among cities ○ = Bottom 1/3 among cities * = data unavailable

Share the road/public safety campaign; info on bicycling in driver's manual; driver's test questions on bicycling; state-sponsored ride to promote bicycling/activity; bicycling enforcement as a policy academy requirement; bicycling enforcement in police continuing education; and existence of an annual statewide bike/ped conference.

Those counted for cities include: Adult and youth bicycle education courses; Bike to Work Day events; open streets initiative; city-sponsored ride to promote bicycling/activity; public bike share program. **Data Source:** State and city surveys

Advocacy Capacity: This ranking is based on the 2010 per capita revenue of Alliance bicycling and walking advocacy organizations serving cities/states. Only statewide organizations are included for states and only organizations with a focus on serving a study area city are included for cities. Cities and states without dedicated Alliance advocacy organizations are marked by an empty circle. **Data Source:** Organization surveys (2)

Notes: (1) Because many states and cities have the same number of policies, policy rankings are not divided into even thirds. For states, those with more than 10 of the 14 policies considered are indicated with full circles; those with 8-10 policies are indicated with a half circle, and those with fewer than 8 policies are indicated with an empty circle. For cities, those with 10 or more of the 13 policies considered are indicated with full circles; those with 6-9 policies are indicated with a half circle, and those with 5 or fewer policies are indicated with an empty circle. (2) Because many states and cities have the same number of education and encouragement initiatives, these rankings are not divided into even thirds. For states, those with 6-7 of the 7 initiatives considered are indicated with full circles; those with 4-5 initiatives are indicated with a half circle, and those with 3 or fewer initiatives are indicated with an empty circle. For cities, those with 5-6 of the 6 initiatives considered are indicated with full circles; those with 3-4 initiatives are indicated with a half circle, and those with 2 or fewer initiatives are indicated with an empty circle. (3) These rankings are based on surveys of Alliance bicycling and walking advocacy organizations only. Because some cities and states are not served by dedicated Alliance advocacy organizations, for states, the 16 served by advocacy organizations with the greatest capacity are marked with a full circle, the 15 remaining states served by advocacy organizations are marked with half circles, and the remaining states not served by statewide Alliance advocacy organizations are indicated with empty circles. For cities the 15 served by advocacy organizations with the greatest capacity are marked with a full circle, the 14 remaining cities served by advocacy organizations are marked with half circles, and the remaining cities not served by dedicated Alliance advocacy organizations are indicated with empty circles.

City	Mode Share	Safety	Funding	Policy (1)	Education/Encouragement (2)	Advocacy Capacity (3)
Albuquerque	◐	◐	●	●	◐	*
Arlington, TX	○	◐	○	●	◐	○
Atlanta	◐	◐	●	◐	◐	●
Austin	◐	◐	◐	◐	◐	◐
Baltimore	●	●	○	◐	◐	○
Boston	●	●	○	○	●	●
Charlotte	○	○	◐	●	◐	◐
Chicago	●	●	○	◐	●	●
Cleveland	◐	●	○	*	*	◐
Colorado Springs	◐	●	○	○	◐	○
Columbus	◐	◐	○	●	◐	●
Dallas	○	○	●	◐	◐	◐
Denver	●	◐	◐	●	●	◐
Detroit	◐	○	◐	*	*	○
El Paso	○	○	◐	◐	○	○
Fort Worth	○	○	○	○	○	◐
Fresno	◐	○	◐	●	◐	○
Honolulu	●	●	○	◐	●	●
Houston	○	○	◐	◐	◐	◐
Indianapolis	○	◐	◐	*	*	●
Jacksonville	○	○	◐	●	◐	○
Kansas City, MO	○	○	●	●	●	◐
Las Vegas	○	◐	○	◐	○	*
Long Beach	◐	◐	◐	○	●	◐
Los Angeles	◐	◐	○	◐	◐	◐
Louisville	◐	○	○	◐	●	◐
Memphis	○	○	◐	◐	◐	◐
Mesa	◐	◐	○	○	◐	○
Miami	◐	○	●	◐	◐	◐
Milwaukee	●	●	◐	◐	◐	◐
Minneapolis	●	●	●	●	●	●
Nashville	○	○	●	●	◐	○
New Orleans	●	◐	●	○	○	◐
New York	●	●	○	◐	●	●
Oakland	●	●	●	●	●	●
Oklahoma City	○	○	*	◐	○	○
Omaha	◐	●	●	◐	◐	*
Philadelphia	●	●	●	●	◐	●
Phoenix	○	○	◐	◐	◐	○
Portland, OR	●	●	●	●	●	●
Raleigh	◐	◐	●	◐	○	○
Sacramento	●	●	●	●	◐	●
San Antonio	○	○	◐	○	◐	○
San Diego	◐	◐	●	◐	◐	◐
San Francisco	●	●	◐	●	●	●
San Jose	◐	◐	◐	●	●	*
Seattle	●	●	◐	●	◐	●
Tucson	●	◐	●	◐	◐	*
Tulsa	○	○	○	○	◐	●
Virginia Beach	○	◐	○	●	○	○
Washington, DC	●	●	●	●	●	*
Find the data (page)	45,47	59,62	87	69, 73	114	140

Infrastructure

City surveys examined current and planned bicycle and pedestrian infrastructure in order to benchmark the progress communities are making. Specifically, cities reported miles of bike lanes, bicycle routes, and multi-use paths. On average, cities have 1.8 miles of bicycle facilities (bike lanes, multi-use paths, and signed bicycle routes) per square mile—a 29% increase since the 2010 Benchmarking Report.

While implementation of innovative facilities such as bicycle boulevards and colored bike lanes is low, surveys indicated that there are new projects currently being implemented or in the process of approval. The number of cities that report having implemented innovative facilities has increased significantly in the last two years. Seventy-three percent of cities now report having implemented sharrows, or shared lane markings.

Bike-Transit Integration

Bike-transit integration has proved to be a vital aspect of effective bicycle systems. The report analyzes responses from city and state surveys, as well as American Public Transportation Association (APTA) data, to see how well cities are integrating bicycle systems with transit. Forty-four cities report that 100% of their bus fleet have bicycle racks, a 19% increase over the past two years. Major U.S. cities report an average of 2.5 bicycle parking spaces at bus stops for every 10,000 residents.

Education and Encouragement

Education and encouragement programs at the state and city level are effective ways to inform the public and promote bicycling and walking. Infor-

mation from state and city surveys and the National Center for Safe Routes to School illustrates the growth in bicycle and pedestrian education in communities. National Walk and Bike to School Day is a popular encouragement activity with growing school participation nationwide.

Thirty-eight cities report having youth bicycle education courses and 41 have adult courses. Youth education is a vital area of outreach because it has the potential to influence the habits of the next generation. The number of youth who participate in bicycle education courses in cities increased by 31% from two years ago. Surveys indicate a 40% increase in adult participation levels for bicycle educational courses over the last two years.

League of American Bicyclists' data indicate that almost all states (49) have information on bicycling in their state driver's manual, yet just 32 states have questions on bicycling on their state driver's exam. The majority of states (38) have a "Share the Road" or similar public safety campaign. Seventeen states report sponsoring a statewide ride to promote bicycling or physical activity.

The Alliance also collected data on professional education regarding bicycling and walking. Overall, these efforts are growing among states, but there is still great room for improvement. Only 20 states have bicycle enforcement as a police academy requirement. And, just 25 states report having hosted a statewide bicycle and pedestrian conference.

Cities were also surveyed on encouragement activities including presence of and participation levels in Bike to Work

Day events, open street/ciclovía initiatives, and city-sponsored bicycle rides. Bike to Work Day is the most common encouragement event with 43 cities participating with an average of one participant for every 286 adults. Thirty-two cities sponsor rides to promote bicycling or physical activity with an average of one participant for every 350 residents. Twenty-one cities have open street (car-free or ciclovía) initiatives with an average of one participant for every 37 residents.

Cycling and Walking Advocacy

Advocacy organizations have the potential to influence bicycling and walking in the communities they serve

by advocating for and winning new policies, funding, infrastructure, and programs. The number of Alliance state and local bicycle and pedestrian advocacy organizations has been increasing steadily since the Alliance was founded in 1996. This report measures organization capacity of Alliance member organizations and sets standards for membership, revenue, staffing, and media exposure. Results from Alliance organization surveys vary widely because of the great variation in maturity and operations of these organizations as well as the communities they serve. Some organizations in this report are decades old while others were founded not long before these surveys were collected.

Photo by Dan Burden, Walkable and Livable Communities Institute



Surveys indicate that organizations serving cities earn significantly more per capita than their statewide counterparts. Local organizations earn an average of \$0.15 per resident served while statewide organizations earn just \$0.03 per resident. In general, organization revenue is diversified, coming from membership and donations, events, fees, grants, contracts, and the bicycle industry. Local Alliance organizations also have much higher per capita membership levels averaging one member per 1,522 residents. Statewide organizations have an average of one member per 4,975 residents. Similarly, statewide organizations operate with an average of 0.4 full-time-equivalent staff (FTE) per million residents served. Organizations serving cities average 2.2 FTE staff per million residents.

Factors Influencing Bicycling and Walking

Analysis in this report shows several positive relationships between bicycling and walking rates and safety, advocacy capacity, density, and car ownership. While weather does not appear to be a factor that directly influences bicycling levels, density, advocacy capacity, and car ownership are a few factors that appear to influence bicycling and walking trips.

ACS and FARS data indicate a positive correlation between bicycling and walking levels and safety. In line with previous studies, an increase in walking and bicycling levels is strongly related to increased bicyclist and pedestrian safety.

Public Health Benefits

To see how bicycling and walking influence public health, the Alliance

compared public health data to bicycling and walking levels. Data from the Behavioral Risk Factor Surveillance System (BRFSS) and ACS reflect a direct relationship between levels of bicycling and walking and several public health indicators. Data suggest that the risk for such health problems as obesity, diabetes, asthma, and hypertension will decrease with more bicycling and walking. States with lower bicycling and walking levels on average have higher levels of obesity, diabetes, hypertension, and asthma. States with higher levels of bicycling and walking also have a greater percentage of adults who meet the recommended 30-plus minutes of daily physical activity. This suggests that increasing bicycling and walking can help achieve public health goals of increasing physical activity and lowering rates of overweight and obesity.

Economic Benefits

To see how bicycling and walking influence the economic strength of communities, the Alliance surveyed numerous studies and data sources. Evidence suggests that bicycling and walking projects create 11-14 jobs per \$1 million spent, compared to just 7 jobs created per \$1 million spent with highway projects. Surveys show that facilities for bicycling and walking attract tourists, event participants, and business. In addition bicycling and walking are affordable investments that save commuters money and in turn equate to more money available for local economies.

Studies that have performed cost/benefit analysis on bicycling and walking facilities have found that these facilities have significant benefit for public health, traffic congestion, and air quality. The cost benefit ratio of Portland, OR's bicycle investments, looking at just

health and fuel savings, ranged from 3.8-to-1 to 1.3-to-1.

Conclusions

While many state and local communities are making sufficient efforts to promote bicycling and walking, much more work needs to be done. Barriers in staffing and funding remain a consistent limitation to promoting bicycling and walking. Bicycling and walking make up 11.5% of all trips, and 13.5% of traffic fatalities, and yet receive just 1.6% of federal transportation dollars.

The proven environmental, economic, and personal health benefits that bicycling and walking offer are evidence that increasing bicycling and walking levels are in the public good, yet a much greater investment is needed throughout the U.S. This Benchmarking

Report identifies which cities and states are leading the way and provides links to resources (Appendix 5) from these communities.

The Alliance recommends that government officials and advocates take the time to evaluate their efforts to promote bicycling and walking. This report can be used by communities to see how they measure up, to identify role models, and to set new goals. Continued benchmarking and improvements in the availability of data will strengthen the report in the coming years, and lend a better understanding of the factors that influence bicycling and walking. Ultimately, by providing a tool for communities to consistently measure progress, evaluate results, and set new targets, this report will advance efforts for a more bicycle- and pedestrian-friendly America.

Photo by Frank Chan, San Francisco Bicycle Coalition





Photo by woodleywonderworks.com

1: INTRODUCTION

Bicycling and walking are good for public health, good for the environment, good for local economies, and help create vibrant communities. This report shows that bicycling and walking are prudent investments that deliver greater returns, and create more jobs, than investing in motorized transportation. Bicycling and walking are also critical components of a healthy active lifestyle that promises to improve health, help protect against various diseases, reduce stress, and improve overall quality of life. For these reasons, government officials, elected representatives, and the media are taking an increased interest in active transportation.

Since publishing the first biennial Benchmarking Report in 2007, there

have been many significant new efforts, programs, organizations, and policies promoting bicycling and walking in the United States. Since 2007, the Alliance's network has grown from 133 to nearly 200 grassroots bicycling and walking advocacy organizations.

In August 2008, the first public smart bike sharing program in the U.S. was launched in Washington, DC, and subsequent programs have sprung up in Boston, Chicago, Denver, Minneapolis, Nashville, San Antonio, and other cities.

In December 2009 the National Association of City Transportation Officials (NACTO) founded the Cities for Cycling project to document, promote, and implement the world's best bicycle transportation practices in U.S. cities.

The League of American Bicyclists has expanded its Bicycle Friendly Communities Program to make states, businesses, and universities eligible for "bicycle friendly" designation. In April 2011, the Pedestrian and Bicyclist Information Center awarded the inaugural Walk Friendly Community designations to 11 communities.

These efforts are receiving increasing support from people in the public eye. In February 2010, First Lady Michelle Obama launched her Let's Move campaign to reduce childhood obesity within one generation. Helping kids become more physically active is one of the program's main goals. In March 2010, U.S. Secretary of Transportation Ray LaHood came out as a champion of bicycling and walking and introduced a new policy that recommended "increased commitment to and investment in bicycle facilities and walking networks." In 2011, a survey of U.S. mayors revealed that they want more control of federal transportation money and 60% of mayors see bicycle and pedestrian projects as a major priority (Flusche 2011).

Public demand for bikeable and walkable places is also growing. A 2010 survey of 1,025 adults age 18 and older found that nearly half of drivers ages 18-34 are driving less. Nearly two-thirds reported they would drive less if transportation alternatives were more readily available. The cost of owning a car and concern for the environment were among the reasons younger drivers are leaving their cars parked (UPI 2010). A 2011 survey by the National Association of Realtors found that Americans favor walkable mixed-use neighborhoods with 56% of respondents selecting these neighborhoods over ones that require more driving between home, work, and

other destinations (National Association of Realtors 2011).

Momentum is growing for bicycle and walking friendly communities. This report gives a good picture of how the landscape is changing for bicycling and walking. It shows which states and cities are making strides and which are setting the benchmarks. Most importantly, it serves as a tool for officials, advocates, researchers, and the media to track and support continued efforts to increase investment in bicycling, walking, safety, and public health.



Photo courtesy of Transportation for America

Benchmarking Bicycling and Walking

Benchmarking is the method of determining best practices or standards and who sets them. Government officials and bicycle and pedestrian advocates have all wondered at some point how their city or state compares with others. Officials and advocates need data to measure their progress and evaluate their efforts. The Alliance for Biking & Walking's Benchmarking Project collects data from government and national data sources, and through surveys to government officials and advocates. Results are published in this biennial Benchmarking Report to measure progress over time of the most-populous cities and states in regard to bicycling and walking.

Benchmarking helps to show officials and advocates where their city or state measures up and helps them to identify areas most in need of improvement. The ultimate objectives of the Benchmarking Project are to increase the number of people who bicycle and walk and to improve their safety. Through benchmarking, new goals can be set, programs evaluated, and continued progress made toward a bicycle and pedestrian friendly America.

Objectives

Promote Data Collection and Availability

Historically there has been little data available on bicycling and walking that can be compared across states and cit-



ies. Data that have existed are often not easily accessible to officials and advocates. One of the main objectives of the Alliance's Benchmarking Project is to promote data collection and availability. This project collects data from a number of government and national data sources and presents it in a way that is easily accessible to those who need it. Through biennial surveys of states, cities, and advocacy organizations, the Benchmarking Project makes new data available such as miles of infrastructure, staffing levels, and advocacy capacity. These data are not available from any other source, but are crucial to understanding mode share and safety outcomes.

Measure Progress and Evaluate Results

Benchmarking is a necessary step to give communities a true picture of how they compare to other communities, what areas they are excelling in, and where they are falling behind. Most importantly, these data enable advocates and officials to evaluate the results of their efforts. Because the Benchmarking Project is ongoing, states and cities can measure their progress over time and will see the impacts of their efforts. By providing a consistent and objective tool for evaluation, this report allows states and cities to determine what works and what doesn't. Successful models can be emulated and failed models discarded.

Support Efforts to Increase Bicycling and Walking

The ultimate objectives of the Alliance's Benchmarking Project are to support the efforts of officials and advocates to increase bicycling and walking in their communities and improve bicycle and pedestrian safety across the U.S. By comparing bicycling and walking statistics across states and cities, this report highlights and praises efforts of communities who provide models, encourages those making progress, and makes states and cities aware of areas where they need work. The Alliance hopes that this report will be used by communities to set goals for increasing bicycling and walking, plan strategies using best practice models, and evaluate results over time. The Alliance strives to make this project a service and tool for officials and advocates so that they can chart the best course toward more bikeable and walkable communities.

Make the Health Connection

The Centers for Disease Control and Prevention (CDC) has declared obesity an epidemic, and people are now looking more closely at the lifestyle choices that may be to blame. Among the top are unhealthy diet and sedentary lifestyles. Studies demonstrate a link between the built environment and levels of physical activity (Frank et al., 2004; Goldberg 2007; Salems and Handy 2008; TRB 2005). The way communities are designed is inextricably linked to the amount of physical activity their residents average. Where environments are built with bicyclists and pedestrians in mind, more people bicycle and walk. These environments increase opportunities for physical activity and promote healthy lifestyles.

Nearly 40% of all trips are two miles or less and 27% are one mile or less (NHTS). These are trips considered an easily bikeable or walkable distance. Now that people are looking for answers to reversing the obesity epidemic, increasing bicycling and walking is an obvious solution.

Alliance for Biking & Walking has partnered with the CDC for this project in an effort to highlight the connection between healthy lifestyles and bicycling and walking. This report includes data on physical activity, obesity and overweight trends, high blood pressure rates, and diabetes, to illustrate the connection between bicycling and walking levels and these health indicators. Along with illustrating the correlation between bicycling and walking and health, the Alliance hopes to show, over time, that as bicycling and walking levels increase, the obesity epidemic begins to reverse.

Study Area Populations

Rank	State	Population	Rank	City	Population
1	California	36,961,664	1	New York	8,391,881
2	Texas	24,782,302	2	Los Angeles	3,831,880
3	New York	19,541,453	3	Chicago	2,850,502
4	Florida	18,537,969	4	Houston	2,260,918
5	Illinois	12,910,409	5	Phoenix	1,593,660
6	Pennsylvania	12,604,767	6	Philadelphia	1,547,297
7	Ohio	11,542,645	7	San Antonio	1,373,677
8	Michigan	9,969,727	8	San Diego	1,306,228
9	Georgia	9,829,211	9	Dallas	1,299,590
10	North Carolina	9,380,884	10	San Jose	964,679
11	New Jersey	8,707,740	11	Detroit	910,848
12	Virginia	7,882,590	12	San Francisco	815,358
13	Washington	6,664,195	13	Jacksonville	813,518
14	Arizona	6,595,778	14	Indianapolis	807,640
15	Massachusetts	6,593,587	15	Austin	790,593
16	Indiana	6,423,113	16	Columbus	773,021
17	Tennessee	6,296,254	17	Fort Worth	731,588
18	Missouri	5,987,580	18	Charlotte	704,417
19	Maryland	5,699,478	19	Memphis	676,646
20	Wisconsin	5,654,774	20	Boston	645,187
21	Minnesota	5,266,215	21	Baltimore	637,418
22	Colorado	5,024,748	22	El Paso	620,440
23	Alabama	4,708,708	23	Seattle	616,669
24	South Carolina	4,561,242	24	Denver	610,345
25	Louisiana	4,492,076	25	Nashville	605,466
26	Kentucky	4,314,113	26	Milwaukee	605,027
27	Oregon	3,825,657	27	Washington, DC	599,657
28	Oklahoma	3,687,050	28	Las Vegas	567,610
29	Connecticut	3,518,288	29	Portland, OR	566,606
30	Iowa	3,007,857	30	Louisville	566,492
31	Mississippi	2,951,996	31	Oklahoma City	560,226
32	Arkansas	2,889,450	32	Tucson	543,907
33	Kansas	2,818,747	33	Atlanta	540,932
34	Utah	2,784,572	34	Albuquerque	529,216
35	Nevada	2,643,085	35	Kansas City, MO	482,228
36	New Mexico	2,009,671	36	Fresno	479,911
37	West Virginia	1,819,777	37	Mesa	467,178
38	Nebraska	1,796,622	38	Sacramento	466,685
39	Idaho	1,545,801	39	Long Beach	462,594
40	New Hampshire	1,324,575	40	Omaha	454,714
41	Maine	1,318,301	41	Virginia Beach	433,575
42	Hawaii	1,295,178	42	Miami	433,143
43	Rhode Island	1,053,209	43	Cleveland	431,369
44	Montana	974,989	44	Oakland	409,151
45	Delaware	885,122	45	Raleigh	405,197
46	South Dakota	812,383	46	Colorado Springs	399,803
47	Alaska	698,473	47	Tulsa	389,369
48	North Dakota	646,844	48	Minneapolis	385,384
49	Vermont	621,760	49	Arlington	380,072
50	Wyoming	544,270	50	Honolulu	374,658
			51 *	New Orleans	354,850

Source: 2009 ACS Note: * New Orleans is not currently the 51st largest U.S. city but was included in this report for consistency and continuity with the 2007 and 2010 Benchmarking Reports.

Data and illustrations in this report are intended to be used by officials and advocates to argue for bicycling and walking as an important part of the solution to creating healthier communities.

Strengthen the Alliance's Network

Lastly, the Alliance aims to strengthen its network of bicycle and pedestrian advocacy organizations by providing organizations the data they need to evaluate their success, prove results, and gain prominence in their communities. Alliance organizations can show data from this report to their community leaders, government officials, and media to highlight areas in which their community is successful, making progress, and in need of improvements.

Alliance organizations can also use these data to prove that advocacy gets results by showing the link between advocacy capacity and levels of bicycling and walking. This report is a tool for Alliance member organizations to gain prominence and win safe and accessible streets for bicycling and walking in their communities.

Study Areas and Data Collection

50 States / 51 Cities

The Benchmarking Project focuses data collection efforts on the 50 U.S. states, the 50 largest U.S. cities,

and New Orleans⁽¹⁾. The 51 largest cities were chosen for this study because these areas are the largest population areas of U.S. residents. Cities are also generally more densely developed than suburban and rural communities, and so may have greater opportunities for conversion of car trips to bicycling and walking.

National Data Collection

The Project Team identified national and uniform government sources for data in this report whenever possible. National data sources utilized for this report include:

- American Community Survey (ACS) (2005-2009)
- American Public Transportation Association (APTA) (2010)
- Behavioral Risk Factor Surveillance System (BRFSS) (2009)
- Federal Highway Administration's FMIS (FHWA) (2004-2010)
- Fatality Analysis Reporting System (FARS) (2005-2009)
- League of American Bicyclists (LAB) Bicycle Friendly States Program (2011)
- National Center for Safe Routes to School (2011)
- National Complete Streets Coalition (2011)
- National Health Interview Survey (NHIS) (2005)
- National Health and Nutrition Examination Study (NHANES) (2005-2006)
- National Household Travel Survey (2001, 2009)

Note: (1) New Orleans was included in the 2007 Benchmarking Report as a top 50 population city (according to 2005 ACS population data), but experienced dramatic population loss after Hurricane Katrina in 2005. Raleigh, NC, moved up into the top 50 largest cities and was the only new city added for this report in 2010. The project team chose to keep New Orleans in this analysis to maintain consistency in cities reported. Throughout this report we refer to the "51 Largest U.S. Cities" which includes the 50 largest U.S. cities and New Orleans. Throughout this report, the top 51 largest U.S. cities are also referred to as "major" or "largest" U.S. cities.



Photo by Jim Swanson

Primary Benchmarks in This Report

Input Benchmarks	
Policy (Chapter 4)	♦ funding levels (per capita and % of transportation dollars to bicycling and walking) ♦ complete streets policies ♦ goals to increase bicycling and walking ♦ goals to increase safety ♦ bike/ped master plan ♦ bike/ped advisory committee ♦ legislation ♦ infrastructure (existing and planned miles per square mile) ♦ bike-transit integration • bicycle racks on buses • bicycle parking spaces at transit stations (per capita) • bicycle access on rail
Programs (Chapter 5)	♦ adult and youth bicycle education courses participation (per capita) ♦ Bike to Work Day participation (per capita) ♦ open streets (ciclovía) initiatives participation (per capita) ♦ city/state-sponsored bicycle rides participation (per capita) ♦ Walk and Bike to School Day participation (per capita)
Advocacy (Chapter 6)	♦ presence of dedicated bike/ped advocacy organization ♦ capacity indicators of advocacy organization • membership (per capita) • income (per capita) • staff levels (per capita) • contacts (per capita)
Outcome Benchmarks	
Mode share (Chapter 2)	♦ share of commuters ♦ all trips ♦ demographics • age • gender • ethnicity • income
Safety (Chapter 3)	♦ fatalities (number and percent of all traffic fatalities) ♦ risk ♦ disparities in mode share and fatalities ♦ demographics • age
Public health (Chapter 8)	♦ overweight and obesity levels ♦ hypertension (high blood pressure) levels ♦ diabetes levels ♦ asthma levels ♦ physical activity levels

- National Transportation Enhancements Clearinghouse (2011)
- Rails-to-Trails Conservancy (2011)
- Safe Routes to School National Partnership (SRTSNP) State of the State's Report (2011)
- School Transportation News (2011)
- U.S. Census (1990, 2000)
- United States Historical Climatology Network (USHCN)
- Web-based Injury Statistics Query and Reporting System (WISQARS) (2009)

In some cases, data in this report come from individual independent studies. The sources for all data are identified throughout the report with accompanying charts, tables, and graphics. An overview of the national data sources used in this report can be found in Appendix 1 on page 199. Individual studies cited in this report can be referenced in the Bibliography on page 231.

State and City Surveys

Many of the variables this report measures are not currently available from other national sources. In these cases, the project team relied on surveys completed by city and state agencies for data on indicators such as miles of bicycle facilities, city and state education efforts, and policies. The surveys were sent to leaders of Alliance organizations, government officials, and advocates in the 50 states and 51 cities represented in this report in October 2010. Because Alliance advocacy leaders can tap existing relationships with local government officials, they were able to help increase the survey response rate and ensure that finished surveys were as complete as possible.

Surveys were completed by department of transportation staff, metropolitan

planning organization staff, city officials, and Alliance advocacy leaders. In many cases surveys required input from multiple agencies because the requested data were not easily accessible in one place. The project team reached out to survey respondents through March 2011, with the final data for the report coming in early April. All data were entered into the Benchmarking Project's data collection tool, checked for quality control, and analyzed over the next several months. This report relies largely on self-reported data and while the Alliance has made all efforts to verify, the accuracy cannot be guaranteed.

Benchmarks in This Report

Bicycling and walking mode share
(percent of all trips and percent of trips

to work) and safety are the two primary outcome benchmarks of this project. Because our ultimate goals are to increase bicycling and walking, and improve bicyclist and pedestrian safety, these are the ultimate benchmarks to measure the progress of states and cities. We also measure a number of variables (called "input benchmarks" here) which we believe, and research has shown, influence levels of bicycling, walking, and safety. Input benchmarks are the factors that affect the outcome benchmarks. Policies, programs, and advocacy capacity are the three primary areas measured in this report. While likely no single policy or program measured here is solely responsible for bicycling and walking levels and safety, a number of them combined may shape mode share and safety levels.

This report includes additional data on factors that may influence bicycling and walking including weather, residential



density, and levels of car ownership. This report also includes data on public health, an outcome benchmark of this project.

Using This Report

The Benchmarking Project is intended as a resource for government officials, bicycle and pedestrian advocates, researchers, and the media searching for comparable data and means to measure progress. We encourage you to search this report for your city or state to see how you compare to others. To make data easy to find, this report orders all data tables alphabetically by city or state. Charts and graphs are ordered by benchmark in order to most clearly see how states and cities compare with each other. **Here are a few additional tips for using this report:**

1. **See where you measure up:** Review the report for your city or state. See how your city/state compares to others. Are you below or above the average for other cities/states? Note where you are leading and where you are behind.
2. **Connect with the media:** Consider issuing a press release or talking with the media about this report. Discuss how your state or city stacks up against others in bicycling and walking levels, safety, and funding. Highlight any areas where you are leading and opportunities for improvement. Use the data to support the work you are doing to promote bicycling and walking locally.
3. **Evaluate your efforts:** Think about where you have been focusing your efforts toward increasing bicycling and walking and safety. Are these efforts working? Look for trends in the data in this report. Look for benchmarks set by cities and states that are leading in the area you are working in.
4. **Set new goals:** Use the data in this report to set new goals and refocus your efforts if needed. There are examples in this report of significant improvements in just a few short years. You will find which cities and states are leading in funding, safety, facilities, and other areas and will also see what the national average and averages for major U.S. cities are. Use these benchmarks to set goals for your city/state.
5. **Use it as a reference book:** The Alliance has heard from a number of government officials and advocates that the Benchmarking Report is a publication they reference frequently in their work. Keep this report on your office bookshelf in an accessible location or digital format. Use it when you are contacted by the media for statistics in your community, or when you need facts for a presentation or paper you are preparing. Use these data to support your work promoting bicycling and walking in your state or city.
6. **Share it:** Purchase extra hard copies of the report to give to your local elected and agency officials, organization leaders, and others who can use it. The report can be a great reason to have a meeting, talk about the current status, and improvements you can mutually strive for. It is always best to deliver the report in person. Also share the link to the Benchmarking Project Website with members, allies, and funders."

If you have questions about the data in this report, would like to request additional data from the Benchmarking Project, have feedback for our team, or other questions or inquiries, please don't hesitate to contact us at benchmarking@PeoplePoweredMovement.org.



Photo courtesy of Ernesto De Quesada

2: LEVELS OF BICYCLING AND WALKING

How Many People Bicycle and Walk?

The question of how many people in a given area bicycle and walk, and what percentage of trips bicycling and walking account for, is arguably the most important question for advocates and officials. Bicycling and walking levels are the ultimate outcome benchmarks of all efforts to promote bicycling and walking. These figures show communities if they are gaining or losing ground in their efforts to convert more trips to active transportation. Unfortunately, accurate and comparable data on bicycling and walking levels are still very limited⁽¹⁾.

Trip Data for This Report

This report relied on the most consistent and dependable source of data on levels of bicycling and walking available: the American Community Survey (ACS). The ACS is an annual survey which provides yearly estimates on the share of workers who usually commute by bicycle or foot. ACS data are available as 1-year estimates, 3-year estimates, and 5-year estimates. Five-year estimates provide the greatest accuracy, and 1-year estimates provide the most current data. In this report, 3-year estimates were used when comparing states and cities to provide a current, yet more accurate picture of levels of biking and walking. One-year estimates are used for national averages only. This report also includes the estimated bi-

⁽¹⁾ For a discussion of the challenges with determining accurate levels of bicycling and walking, see Appendix 3, page 202. Appendix 3 also contains a discussion on the differences between the ACS and Census methodologies.

STATE RANKING

Cycling to Work

1. Oregon
2. Montana
3. Idaho
4. Colorado
5. Wyoming
6. California
7. Hawaii
8. Alaska
9. Washington
10. Arizona
11. Minnesota
12. Utah
13. New Mexico
14. Wisconsin
15. Massachusetts
16. Vermont
17. North Dakota
18. Florida
19. Illinois
20. South Dakota
21. Nevada
22. Nebraska
23. Maine
24. New York
25. Pennsylvania
26. Michigan
27. Iowa
28. Indiana
29. Kansas
30. Delaware
31. Louisiana
32. New Hampshire
33. Virginia
34. Ohio
35. New Jersey
36. Rhode Island
37. Maryland
38. Connecticut
39. Texas
40. South Carolina
41. North Carolina
42. Oklahoma
43. Kentucky
44. Missouri
45. Georgia
46. Mississippi
47. West Virginia
48. Tennessee
49. Arkansas
50. Alabama

Walking to Work

1. Alaska
2. New York
3. Vermont
4. Montana
5. Hawaii
6. South Dakota
7. Massachusetts
8. Maine
9. Wyoming
10. Pennsylvania
11. Iowa
12. North Dakota
13. Oregon
14. Washington
15. Wisconsin
16. New Jersey
17. New Hampshire
18. Nebraska
19. Illinois
20. Idaho
21. Rhode Island
22. Minnesota
23. Colorado
24. Connecticut
25. Utah
26. West Virginia
27. California
28. Kansas
29. Maryland
30. Delaware
31. New Mexico
32. Kentucky
33. Michigan
34. Nevada
35. Virginia
36. Ohio
37. Indiana
38. Arizona
39. Louisiana
40. Missouri
41. Oklahoma
42. North Carolina
43. South Carolina
44. Mississippi
45. Arkansas
46. Texas
47. Georgia
48. Florida
49. Tennessee
50. Alabama

cycling and walking mode share for all trips from the 2009 National Household Travel Survey (NHTS).

This report looks at the share of commuters who walk or bike to work using data from the 1990 and 2000 decennial Census, and annual ACS between 2005 and 2009, and the most recent 3-year average (2007-2009) and 5-year average (2005-2009) from the ACS. Although work trips account for only 16% of all trips (NHTS 2009), these data provide a glimpse into trends in bicycling and walking levels over the last 19 years.

Findings on Mode Share

The Alliance used 2009 ACS data to determine that nationwide, an average of 3.5% of commuters get to work by bicycle (0.6%) or foot (2.9%). In the major U.S. cities studied here, the share of commuters by bicycle and foot is higher at 5.9% (1.0% bicycling and 4.9% walking). People in major cities are 1.7 times more likely to bicycle to work, and 1.7 times more likely to walk to work, than their counterparts nationwide.

Since the 2010 Benchmarking Report, Oregon remains the state with the highest bicycle to work share at 2.1%. Portland retains the highest share of workers commuting by bicycle—5.5%—among cities in this study. Alabama, Arkansas, and Tennessee rank lowest in bicycle to work commute share with only 0.1% of work trips by bicycle. San Antonio, Oklahoma City, and Dallas

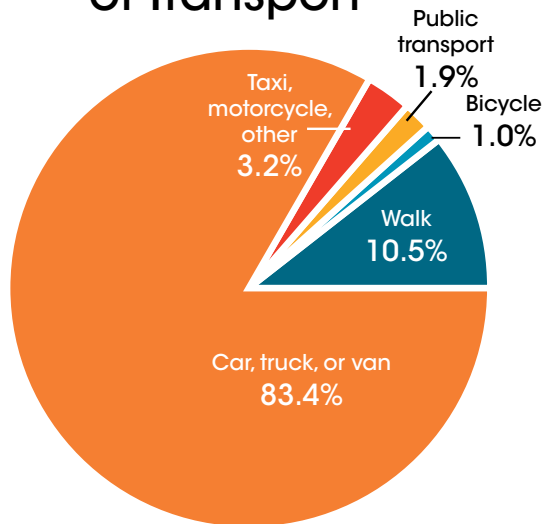
Tables to left: Source: 2007-2009 ACS Notes: This ranking is based on the 3-year average share of commuters who bicycle and walk to work. The state with the greatest share of commuters who bicycle or walk is ranked #1. The 50th position is the state with the least percentage of commuters who bicycle or walk. View these data on pages 45 and 46 of this report..

rank lowest among cities for bicycle work commute share with just 0.1% of work trips by bicycle.

Alaska and Boston remain the state and city with the highest pedestrian commute share (8.0% and 13.9% of all workers commute by foot, respectively). Alabama, with only 1.3% of work trips by foot, ranks lowest among states. Fort Worth has the lowest pedestrian commute share among cities—just 1.2% of work trips by foot.

According to 2009 NHTS estimates the total bicycle mode share for all trip purposes nationwide is 1.0%. In the largest metropolitan areas, 1.1% of all trips are by bicycle.

U.S. Trips by Mode of Transport



Source: NHTS 2009 (Graph above) and ACS 2007-2009 (ranking to right) **Notes:** (ranking to right) This ranking is based on the share of commuters who bicycle and walk to work in cities. The city with the greatest percent of commuters who bicycle or walk is ranked #1. The 51st position is the city with the least percentage of people who commute by bicycle or foot. View this data on pages 45 and 47 of this report. (1) For details and reliability of state and city level NHTS estimates, please see Appendix 3, page 202.

CITY RANKING

Cycling to Work

1. Portland, OR
2. Minneapolis
3. Seattle
4. San Francisco
5. Sacramento
6. Oakland
7. Washington, DC
8. Tucson
9. Denver
10. New Orleans

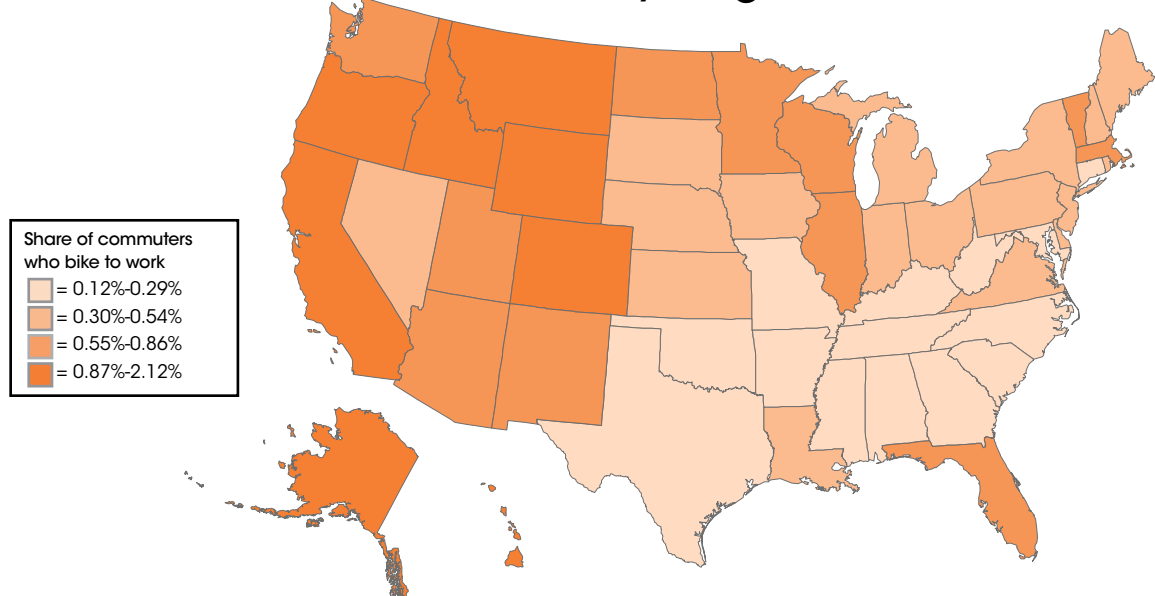
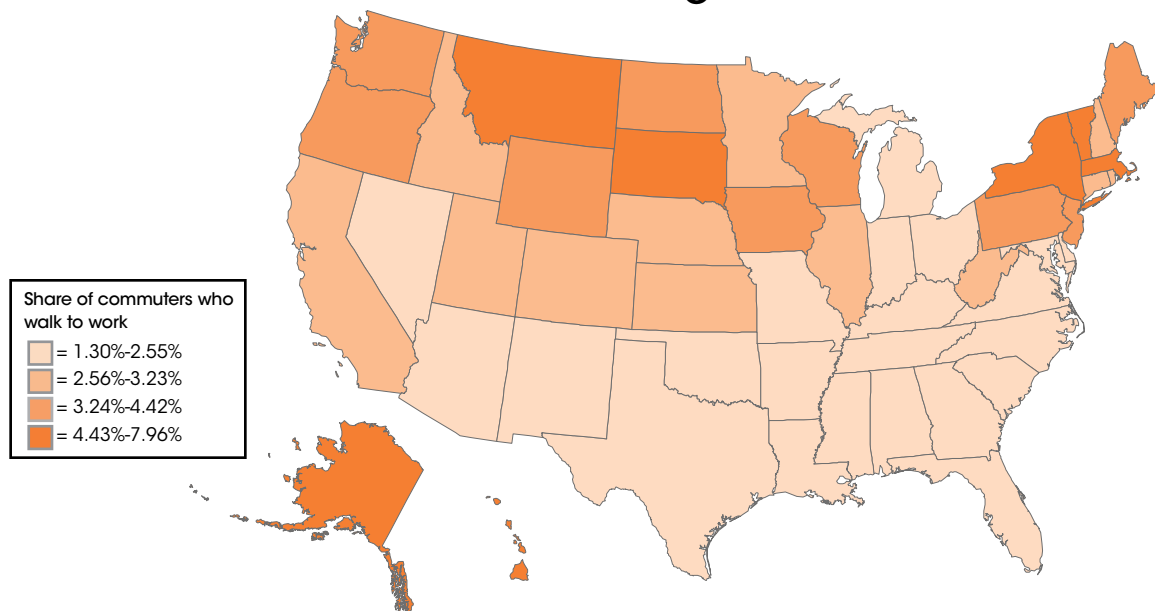
11. Honolulu
12. Philadelphia
13. Boston
14. Albuquerque
15. Austin
16. Chicago
17. Mesa
18. Long Beach
19. San Jose
20. San Diego
21. Los Angeles
22. Milwaukee
23. Atlanta
24. Columbus
25. Phoenix
26. New York
27. Fresno
28. Baltimore
29. Cleveland
30. Colorado Springs
31. Virginia Beach
32. Raleigh
33. Louisville
34. Jacksonville
35. Las Vegas
36. Tulsa
37. Detroit
38. Houston
39. Miami
40. Indianapolis
41. Nashville
42. Kansas City, MO
43. Arlington, TX
44. Omaha
45. El Paso
46. Memphis
47. Fort Worth
48. Charlotte
49. Dallas
50. Oklahoma City
51. San Antonio

Walking to Work

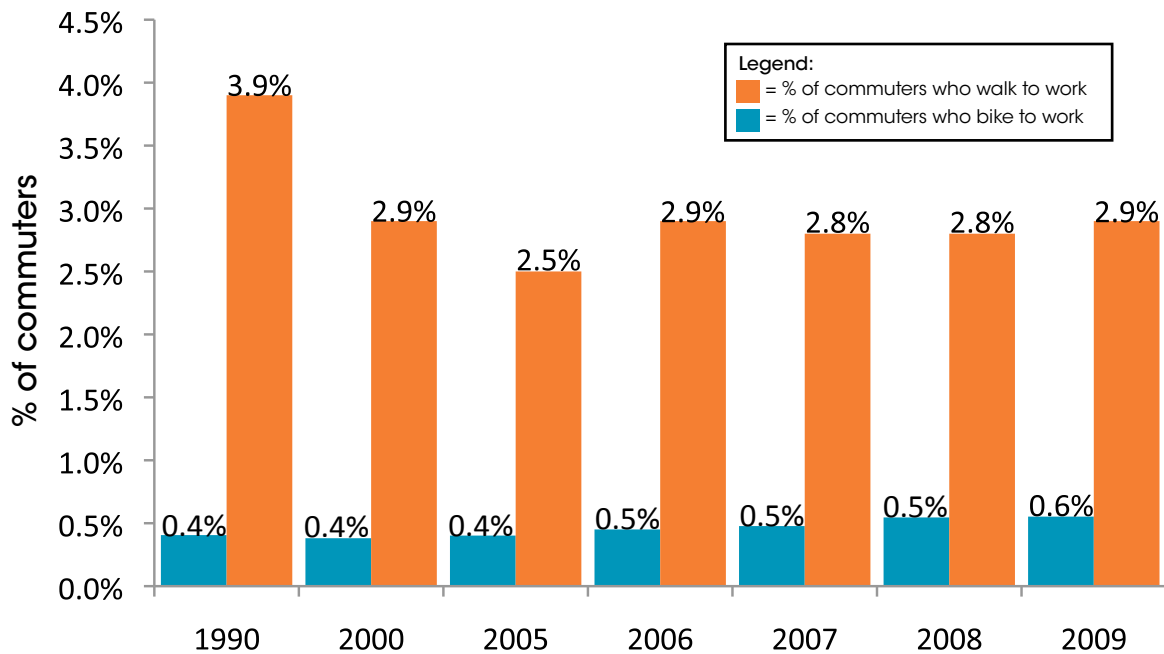
1. Boston
2. Washington, DC
3. New York
4. San Francisco
5. Seattle
6. Philadelphia
7. Honolulu
8. Baltimore
9. Minneapolis
10. New Orleans

11. Chicago
12. Portland, OR
13. Milwaukee
14. Cleveland
15. Oakland
16. Atlanta
17. Denver
18. Miami
19. Tucson
20. Los Angeles
21. Detroit
22. Sacramento
23. Long Beach
24. San Diego
25. Omaha
26. Columbus
27. Raleigh
28. Colorado Springs
29. Louisville
30. Tulsa
31. Houston
32. El Paso
33. Kansas City, MO
34. Indianapolis
35. Virginia Beach
36. Albuquerque
37. Austin
38. Memphis
39. Las Vegas
40. San Antonio
41. Fresno
42. Charlotte
43. Mesa
44. San Jose
45. Arlington, TX
46. Phoenix
47. Dallas
48. Nashville
49. Jacksonville
50. Oklahoma City
51. Fort Worth

Levels of walking to work increased in all but six states between 2005 and 2009. Levels of bicycling to work increased in all but four states during this time period. The southern U.S. remains the region with the lowest levels of bicycling and walking to work.



Share of Commuters Who Bicycle and Walk 1990-2009



Sources: U.S. Census 1990, 2000; ACS 2005, ACS 2006, ACS 2007, ACS 2008, ACS 2009

NHTS data for 2009 show that nationally 10.5% of all trips are by foot. This is up 18% from the 2001 level of 8.9%. Rates of walking in major metropolitan areas ("cities") are even greater. NHTS estimates that 12.7% of all city trips are by foot. This is up 15% from the 2001 level of 11% of all city trips.

The 2009 NHTS also asked respondents how many times they took a trip by bicycle or foot in the last week. Results indicate that 13% of people take at least one bicycle trip per week and 68% of people take at least one walking trip per week. These amount to over 4 billion bicycle trips and nearly 41 billion walking trips in 2009 in the United States.

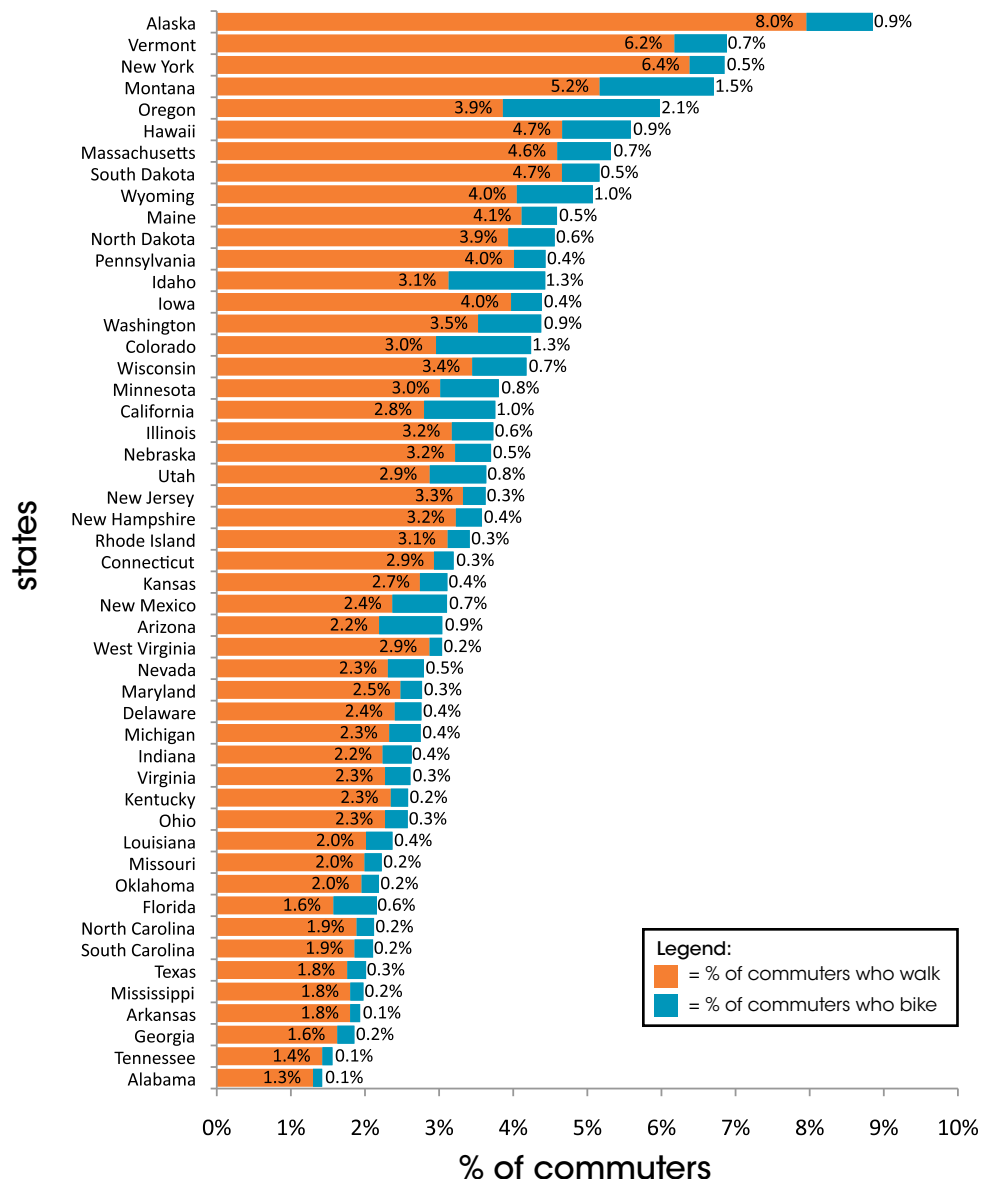
Trends in Bicycling and Walking Levels

The Alliance looked at data from the 1990 and 2000 decennial Census and annual American Community Surveys from 2005 through 2009 to examine trends in the share of commuters who bicycle or walk to work daily over the last two decades. (Find additional data on bicycling and walking levels over time in Appendix 4, page 205.)

The number of people who bicycle to work has increased steadily, rising 64% between 1990 and 2009 from 466,856 to 765,703 people who bicycle to work nationwide. The share of commuters who bicycle to work rose from 0.4% nationwide in 1990 and 2000 to 0.6% in 2009.

(Continued page 41)

Share of Commuters Who Walk and Bicycle in 50 States

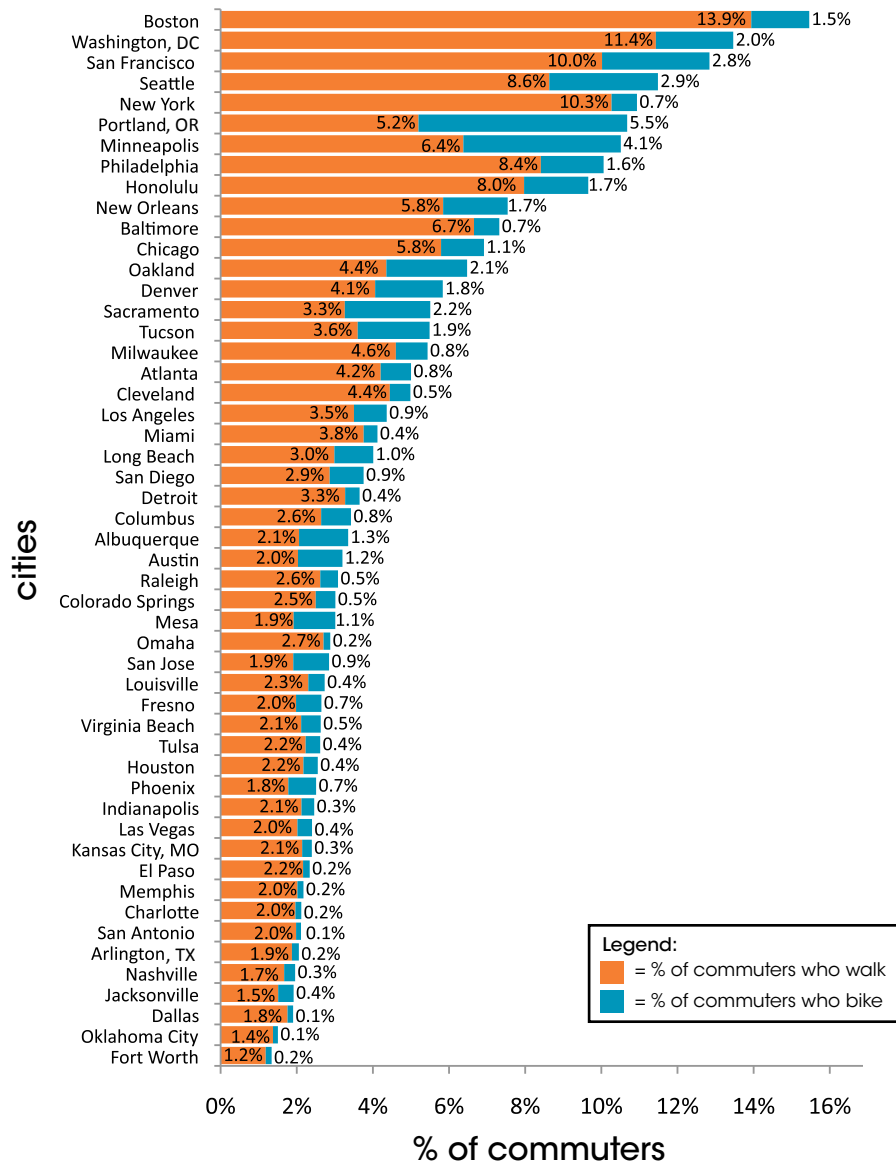


Source: 2007-2009 ACS (3-year average) Note: For a discussion of the challenges with determining accurate levels of bicycling and walking, see Appendix 3.

**Alaska leads states
for bicycle + walk to
work mode share.**

Alaska leads Vermont, New York, and Montana as the state with the highest percentage of commuters who bike or walk—8.9% of all commuters. Alabama and Tennessee rank lowest among states with 1.4% and 1.5% bicycle and walk to work mode share, respectively.

Share of Commuters Who Walk and Bicycle in Largest U.S. Cities



Source: 2007-2009 ACS (3-year average) Note: For a discussion of the challenges with determining accurate levels of bicycling and walking, see Appendix 3.

Bicycling and walking mode share is significantly higher in cities. On average 5.8% of commuters in the largest U.S. cities bike or walk to work. Boston (15.4%) leads Washington, DC (13.4%), San Francisco (12.8%), and Seattle (11.5%) as the cities with the highest rate of bicycling and walking to work.

Boston ranks top for bicycle + walk to work mode share.

Looking Outside the Borders

Bicycling and Walking Levels and Demographics

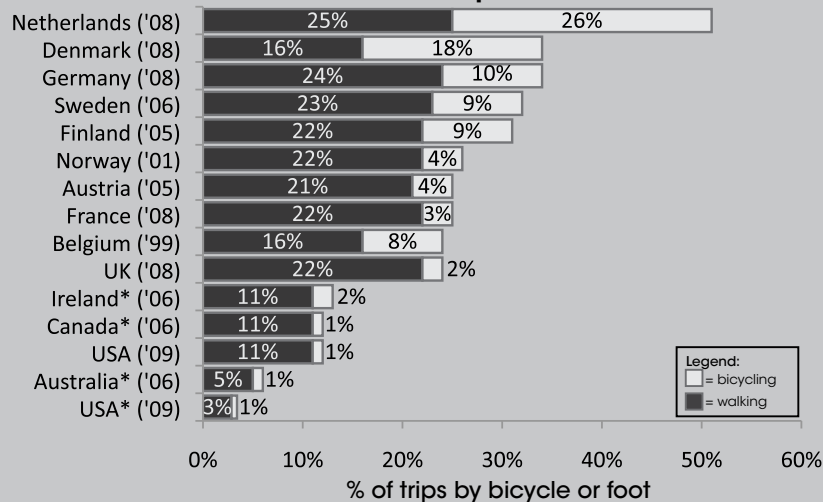
In our efforts to increase bicycling and walking in the United States, it is crucial to learn from the successful policies implemented in other countries, which have far higher levels of walking and bicycling as well as much better safety. A recent study by Pucher and Buehler (2010; chart this page) found that walking and cycling accounted for only about 12% of all trips in the United States in 2009, which is similar to levels in Ireland and Canada, but only about a third as much walking and cycling as many European countries.

With over a third of their trips by walking and cycling, countries like Sweden, Germany, Denmark, and the Netherlands set the standard for active travel in affluent countries with high levels of car ownership. With their bike mode shares of 9%-26%, the same four countries have roughly 10 times as much

cycling as the U.S. Similarly, the U.S. has only about half as much walking as most European countries. The variation among countries is confirmed by large differences among cities in active travel rates, with American cities lagging far behind European cities (Pucher and Buehler 2008; chart page 37). In most large U.S. cities, the bicycle share of trips is less than 1%. Portland, Minneapolis, and Seattle have the highest bicycle to work share among cities: 5.5%, 4.1%, and 2.9%, respectively. By comparison, many cities in Germany, Denmark, and the Netherlands have bicycle trip shares over 10%.

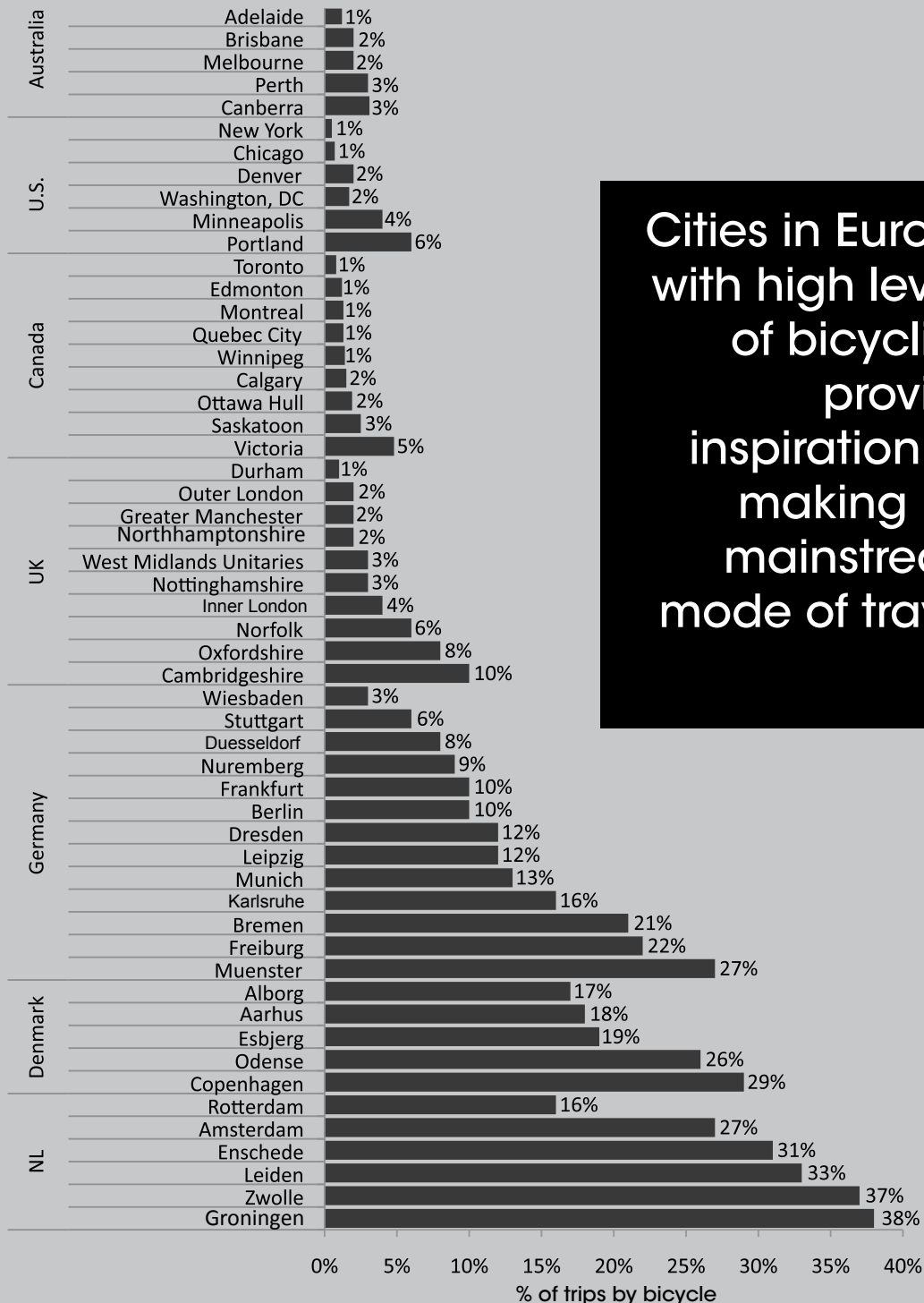
Examining bicycling and walking levels by trip distance shows that in the U.S., 38% of trips shorter than 1.6 miles (2.5 km) were by walking or cycling in 2009. For the same trip distance, the percent of short trips by walking or cycling

Bike and Walk Share of Daily Trips in the USA, Canada, Australia, and 11 European Countries



Source: J. Pucher and R. Buehler, 2010. "Walking and Cycling for Healthy Cities," *Built Environment* 36(5), pp. 391–414. Note: * denotes for the worktrip only, while other country surveys are for all trip purposes.

Bicycle Share of Trips in 55 Cities in the U.S., Canada, Australia, UK, Germany, Denmark, and the Netherlands



Cities in Europe
with high levels
of bicycling
provide
inspiration for
making it a
mainstream
mode of travel.

Source: J. Pucher and R. Buehler, 2008. "Making Cycling Irresistible: Lessons from the Netherlands, Denmark, and Germany," *Transport Reviews* 28(4), pp. 495–528. Note: UK data are for counties.

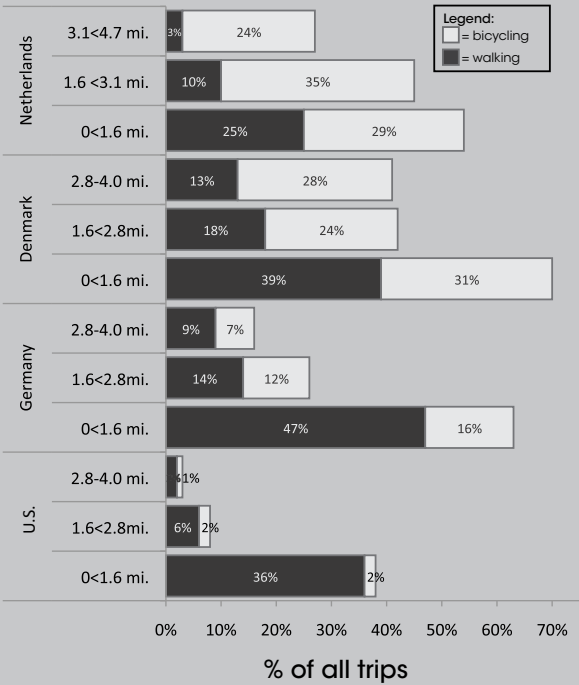


Photo by Kate McCarthy

was 63% in Germany, 70% in Denmark, and 54% in the Netherlands (see chart this page). The variation between the U.S. and other countries is greatest when looking at longer trip distances. Bicycling and walking account for only 3% of trips of 2.8 to 4 miles (4.5-6.5 km) in the U.S., compared to 16% in Germany, 21% in Denmark, and 27% in the Netherlands. Thus, at every trip distance, walking and cycling rates are much higher in northern Europe than in the U.S.

Perhaps the most striking differences among countries in walking and cycling rates are by age group. In Germany,

Bicycling and Walking Levels by Trip Distance



Source: J. Pucher and R. Buehler, 2010 "Walking and Cycling for Healthy Cities," *Built Environment* 36(5), pp. 391–414. URL link: http://policy.rutgers.edu/faculty/pucher/BuiltEnvironment_WalkBike_10Dec2010.pdf

Denmark, and the Netherlands, a high proportion of trips in all age categories are by walking or cycling (see chart page 39). In the U.S., only 18% of trips by children are made by walking or cycling, compared to 43% in Germany, 51% in Denmark, and 64% in the Netherlands. Similarly, only 10% of trips by American elderly are by walking or cycling, compared to 43% in Germany, 51% in Denmark, and 64% in the Netherlands. The much higher levels of walking and cycling in northern Europe provide important physical activity, mobility, and independence for all age groups, while children and seniors in the U.S. are often dependent on their families, neighbors, and friends for many trips they need to make.

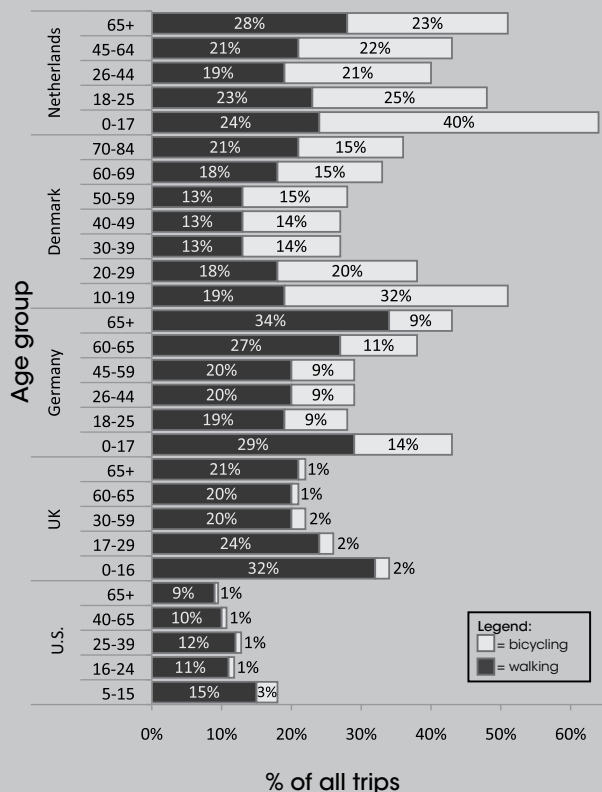
Trends in bicycling and walking levels over the last 35 years vary greatly among

countries (see chart page 40). France and the UK, for example, suffered dramatic falls in walking and cycling, with roughly 50% declines in both countries. By comparison, walking and cycling levels have been roughly stable in the Netherlands, Denmark, and Germany, with declines in walking partly offset by increases in cycling. The much smaller declines in active transport in the Netherlands, Denmark, and Germany are due to far more car-restrictive policies in those countries since the 1970s, combined with a wide range of measures to encourage more walking and cycling. Car-restrictive measures have been far less common in France and the UK, and those two countries have also done less to promote walking and cycling through infrastructure, programs, and policies (Pucher and Buehler 2010). In ad-

dition, suburban and exurban sprawl has been more extensive in France and the UK than in the Netherlands, Denmark, and Germany.

It is more difficult to gauge walking and cycling trends in the U.S. because there was an important change in the national travel survey methodology in 2001 that raised the walk mode share by capturing previously unreported walk trips. The survey results in the chart on page 40 suggest slight increases in walking and cycling levels in the USA, but in fact, they have probably declined. For example, the U.S. Census, using a consistent methodology over time, reports a substantial decline in walking and cycling to work: from 7.9% in 1970 to only 3.3% in 2008 (USDOT 2010).

Bicycling and Walking Levels by Age



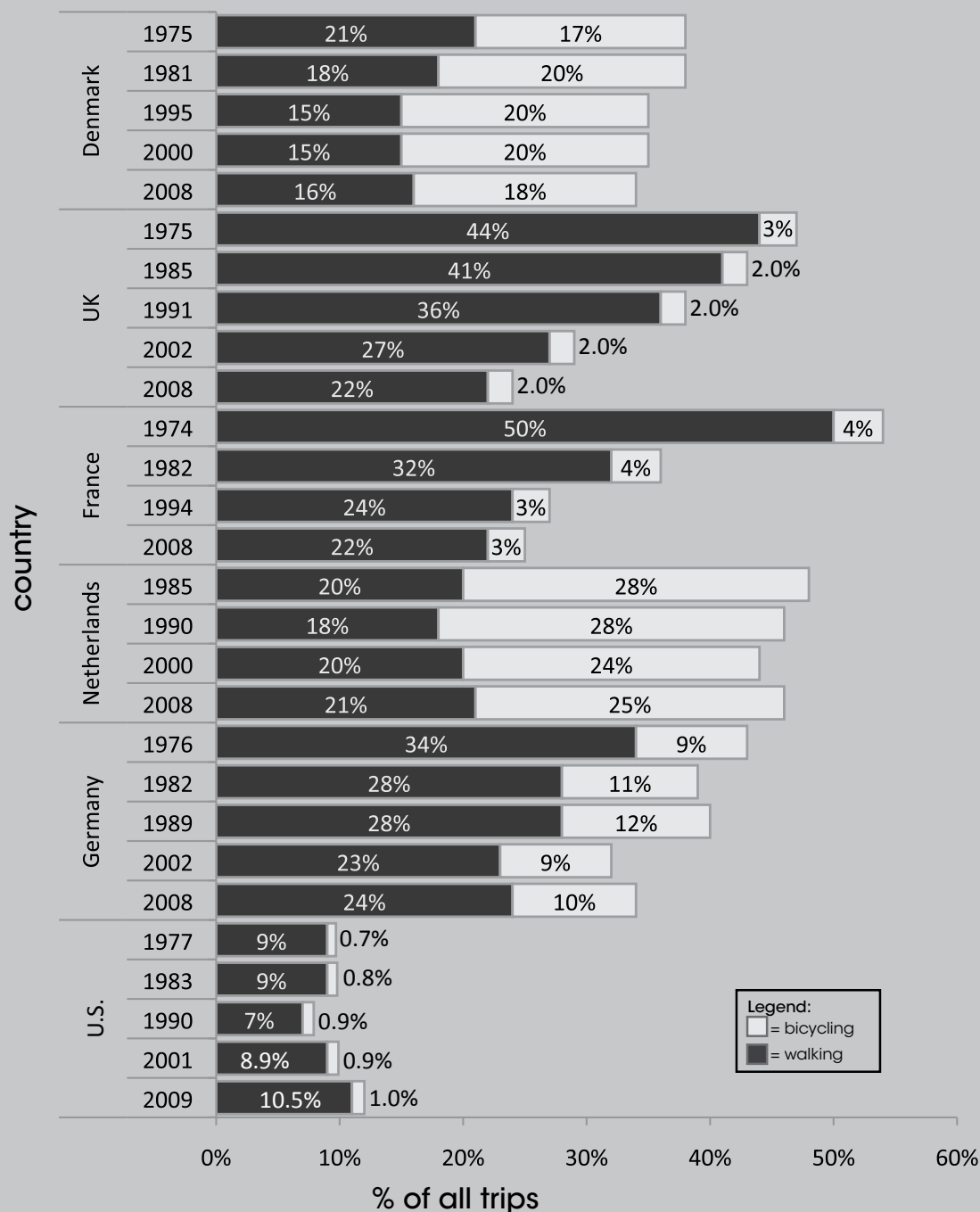
Source: J. Pucher and R. Buehler, 2010. "Walking and Cycling for Healthy Cities," *Built Environment* 36(5), pp. 391–414. URL link: http://policy.rutgers.edu/faculty/pucher/BuiltEnvironment_WalkBike_10Dec2010.pdf Note (above) (1): Methods for NHTS changed in 2001. Thus the increase from 1990 to 2001 may be due to methods. Moreover, the long-term trend may also be skewed because of that.



Photo by Dan Burden, Walkable and Livable Communities Institute

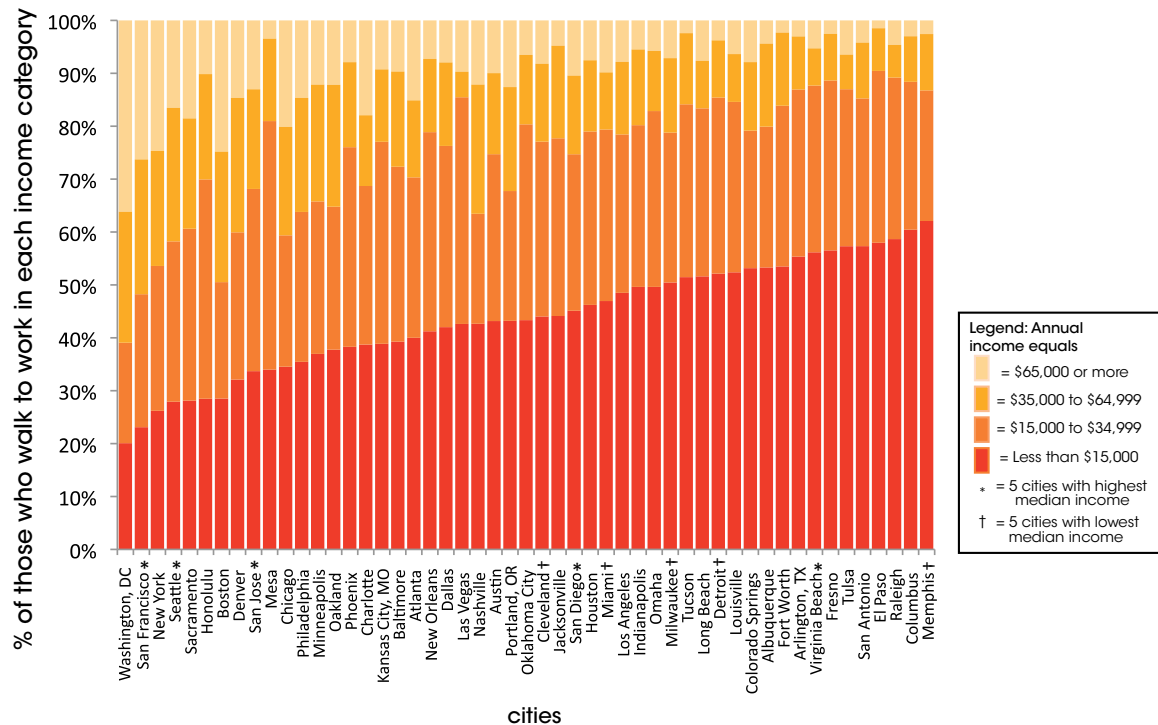
Trends in Cycling and Walking

Share of all daily trips in the U.S., Germany, the Netherlands, France, the UK, and Denmark, 1974-2009



Source: J. Pucher and R. Buehler, 2010. "Walking and Cycling for Healthy Cities," *Built Environment* 36(5), pp. 391-414. URL link: http://policy.rutgers.edu/faculty/pucher/BuiltEnvironment_WalkBike_10Dec2010.pdf. Note: Methods for NHTS (U.S.) changed in 2001. Thus the increase from 1990 to 2001 may be due to methods. Moreover, the long-term trend may also be skewed because of that.

Pedestrian Commuters by Income Classification



During the same time period the number of people who walk to work fell 12% (from roughly 4.5 million people in 1990 to roughly 4.0 million people in 2009). The number of people who walk to work increased by just 5% between 2000 and 2009. The share of commuters who walk to work is now 2.9%, down from 3.9% in 1990. The share of commuters who walk to work has remained relatively stable since 2000.

Who Bicycles and Walks?

Demographic Data

Determining who walks and bicycles is also difficult. Because the ACS counts only commuter trips, trips taken for nonwork purposes, such as those made

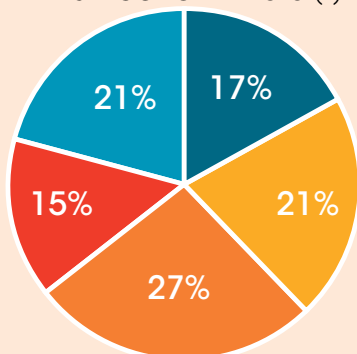
Bicyclist and Pedestrian Mode Share by Income Class

	Household Income					
	Less than \$20,000	\$20,000 to \$39,999	\$40,000 to \$74,999	\$75,000 to \$99,999	\$100,000 and over	All
Bicyclists	1.0%	1.2%	1.0%	0.9%	1.1%	1.1%
Pedestrians	16.3%	10.3%	8.9%	8.9%	10.2%	10.5%

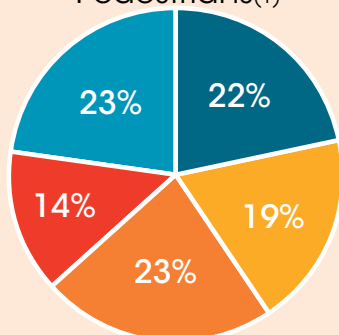
Source: NHTS 2009

A Look at Income

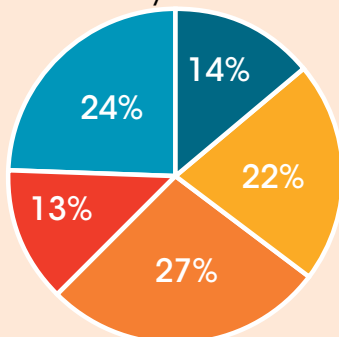
Income
Distribution in U.S.⁽¹⁾



Income
Distribution of
Pedestrians⁽¹⁾



Income
Distribution of
Bicyclists



Legend: Total earnings =
 ■ = Less than \$20,000
 ■ = \$20,000-\$39,999
 ■ = \$40,000-\$74,999
 ■ = \$75,000-\$99,999
 ■ = \$100,000+

Source: NHTS 2009 Note: (1)
Numbers round up and so
appear to add to 101%.

by children, for recreational purposes, or in combination with other modes of transportation are left out. Many local trip count efforts include demographic survey questions (including some referenced in Appendix 6 of this report). However, because there is no standardized format used for these local surveys, the Alliance relied on ACS and NHTS data for demographic information.

Bicyclist and Pedestrian Income

There is almost no variation in the bicycle mode share by income class. Data from the 2009 NHTS show that bicycling mode share is similar for all income classes. However, a closer look at data by trip purpose reveals the percent of bicycle trips for recreational purposes rises with increasing income, from 27% to 41%, while the percent of utilitarian trips falls correspondingly with increased income. Regardless of the reason for bicycle trips, these data show that bicycling levels are roughly evenly distributed among all income classes.

While bicycling is distributed evenly among all income groups, walking to work is mainly by lower income workers. Data from ACS reveal that nearly half of people who walk to work earn less than \$15,000 per year. More than two-thirds of people who walk to work, on average, earn below \$35,000 a year. Washington, DC, San Francisco, and New York City have the most even income distribution among people who walk to work, with all income groups well represented. Memphis and Columbus have the least equal distribution with walking concentrated mostly among low-income groups. The difference in average median income among cities could also account for some variation and should be considered with these data.

Bicyclist and Pedestrian Gender

The gap between men and women is much wider among bicyclists than pedestrians. Na-

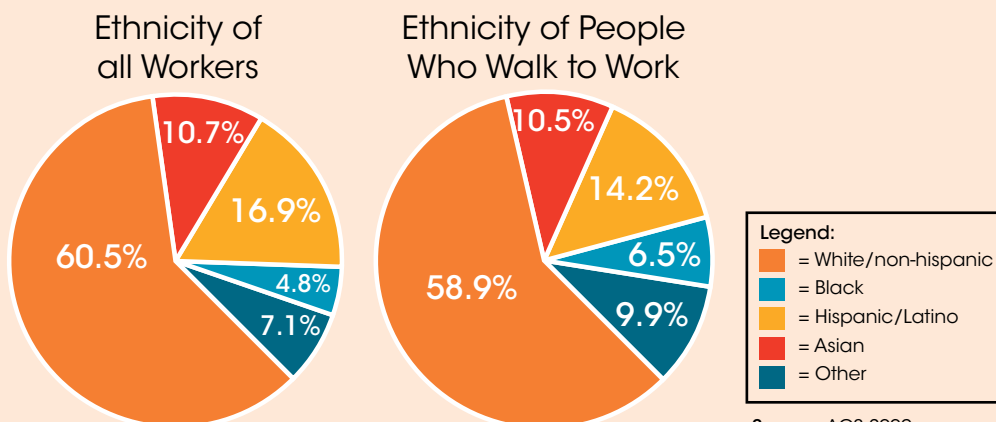


tionwide, just 27% of bicycle commuters are women (up from 23% in 2007). Just 24% of all bicycle trips are female according to the 2009 NHTS. Men make up 73% of bicycle commuters and 54% of pedestrian commuters. Walking is more even between the sexes. Men comprise 49% of the population and the same percentage of all walking trips.

Massachusetts and Wyoming are the only states where women walk to work at slightly higher rates than men. Men bicycle to work at higher rates than women in all states, though the gap varies among states. Montana has the smallest gap among men (66%) and women (34%) bicyclists. Delaware has the largest gap between men (91%) and women (9%) bicyclists.

The gap between men and women also varies largely among major U.S. cities. Again, most cities have relatively small gaps between levels of men and women who walk to work. In roughly 1/6 of cities surveyed, women walk to work at slightly higher rates. Philadelphia has the greatest percentage of pedestrian commuters who are women—55%. Mesa has the least percentage of pedestrian commuters who are women—30%.

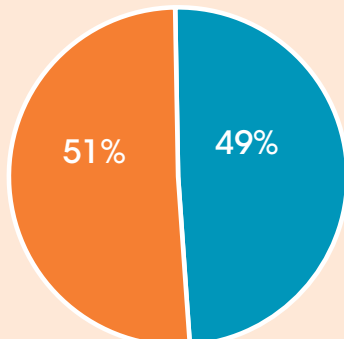
A Look at Ethnicity



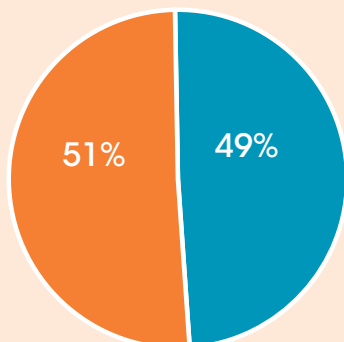
Source: ACS 2009

A Look at Gender

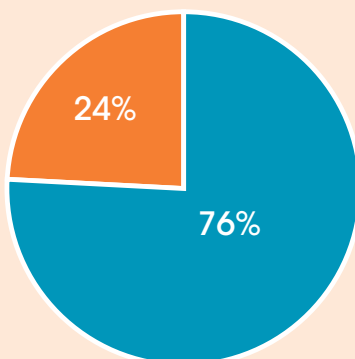
Gender Distribution in U.S.



Walk Trips by Gender



Bike Trips by Gender



Legend:
■ = Female
■ = Male

Source: NHTS 2009



Photo courtesy of Transportation for America

On average, men make up 76% of bicycle commuters in the U.S. and 72% in major U.S. cities. According to ACS data, the vast majority of bicycle commuters in Omaha, El Paso, and Dallas are male, making these the cities with the greatest gender divide among bicyclists. Because of low sample sizes, it is possible that there are more female commuters in these cities, but it is not reflected in the data.

Bicyclist and Pedestrian Ethnicity

ACS data reveal a fairly even distribution among bicyclists and pedestrians in regard to ethnicity. Asians, hispanics, and white/non-hispanic groups are slightly less likely to walk to work. African Americans/blacks are 35% more likely to walk to work, comprising 4.8% of the U.S. workers, but accounting for 6.5% of pedestrian commuters. "Other ethnicities" which include American Indian, Alaska Native, Native Hawaiian, Other Asian Pacific Islander, and other minority ethnicities alone are also more likely to walk to

(Continued page 48)

Bicycling to Work in States

States	% of commuters who bike to work	% men	% women
Alabama	0.1%	84%	16%
Alaska	0.9%	68%	32%
Arizona	0.9%	78%	22%
Arkansas	0.1%	74%	26%
California	1.0%	75%	25%
Colorado	1.3%	71%	29%
Connecticut	0.3%	75%	25%
Delaware	0.4%	91%	9%
Florida	0.6%	76%	24%
Georgia	0.2%	76%	24%
Hawaii	0.9%	71%	29%
Idaho	1.3%	70%	30%
Illinois	0.6%	77%	23%
Indiana	0.4%	73%	27%
Iowa	0.4%	71%	29%
Kansas	0.4%	75%	25%
Kentucky	0.2%	68%	32%
Louisiana	0.4%	79%	21%
Maine	0.5%	67%	33%
Maryland	0.3%	80%	20%
Massachusetts	0.7%	72%	28%
Michigan	0.4%	71%	29%
Minnesota	0.8%	70%	30%
Mississippi	0.2%	86%	14%
Missouri	0.2%	79%	21%
Montana	1.5%	66%	34%
Nebraska	0.5%	73%	27%
Nevada	0.5%	81%	19%
New Hampshire	0.4%	74%	26%
New Jersey	0.3%	80%	20%
New Mexico	0.7%	74%	26%
New York	0.5%	77%	23%
North Carolina	0.2%	80%	20%
North Dakota	0.6%	81%	19%
Ohio	0.3%	73%	27%
Oklahoma	0.2%	69%	31%
Oregon	2.1%	67%	33%
Pennsylvania	0.4%	73%	27%
Rhode Island	0.3%	69%	31%
South Carolina	0.2%	79%	21%
South Dakota	0.5%	72%	28%
Tennessee	0.1%	77%	23%
Texas	0.3%	78%	22%
Utah	0.8%	72%	28%
Vermont	0.7%	68%	32%
Virginia	0.3%	72%	28%
Washington	0.9%	70%	30%
West Virginia	0.2%	74%	26%
Wisconsin	0.7%	70%	30%
Wyoming	1.0%	75%	25%
Mean/Average (1)	0.5%	74%	26%
Median	0.4%	74%	26%
High	2.1%	91%	34%
Low	0.1%	66%	9%

Bicycling to Work in Cities

Cities	% of commuters who bike to work	% men	% women
Albuquerque	1.3%	69%	31%
Arlington, TX	0.2%	84%	16%
Atlanta	0.8%	71%	29%
Austin	1.2%	72%	28%
Baltimore	0.7%	77%	23%
Boston	1.5%	67%	33%
Charlotte	0.2%	82%	18%
Chicago	1.1%	76%	24%
Cleveland	0.5%	65%	35%
Colorado Springs	0.5%	74%	26%
Columbus	0.8%	76%	24%
Dallas	0.1%	91%	9%
Denver	1.8%	70%	30%
Detroit	0.4%	78%	22%
El Paso	0.2%	92%	8%
Fort Worth	0.2%	88%	12%
Fresno	0.7%	77%	23%
Honolulu	1.7%	72%	28%
Houston	0.4%	77%	23%
Indianapolis	0.3%	75%	25%
Jacksonville	0.4%	80%	20%
Kansas City, MO	0.3%	63%	37%
Las Vegas	0.4%	89%	11%
Long Beach	1.0%	85%	15%
Los Angeles	0.9%	80%	20%
Louisville	0.4%	62%	38%
Memphis	0.2%	51%	49%
Mesa	1.1%	84%	16%
Miami	0.4%	70%	30%
Milwaukee	0.8%	73%	27%
Minneapolis	4.1%	63%	37%
Nashville	0.3%	78%	22%
New Orleans	1.7%	76%	24%
New York	0.7%	75%	25%
Oakland	2.1%	66%	34%
Oklahoma City	0.1%	81%	19%
Omaha	0.2%	97%	3%
Philadelphia	1.6%	70%	30%
Phoenix	0.7%	84%	16%
Portland, OR	5.5%	62%	38%
Raleigh	0.5%	76%	24%
Sacramento	2.2%	67%	33%
San Antonio	0.1%	81%	19%
San Diego	0.9%	72%	28%
San Francisco	2.8%	70%	30%
San Jose	0.9%	79%	21%
Seattle	2.9%	68%	32%
Tucson	1.9%	71%	29%
Tulsa	0.4%	75%	25%
Virginia Beach	0.5%	53%	47%
Washington, DC	2.0%	67%	33%
Mean/Average (1)	0.9%	72%	28%
Median	0.7%	75%	25%
High	5.5%	97%	49%
Low	0.1%	51%	3%

Source: ACS 2007-2009 (3-year average) Notes: (1) All averages are weighted. (2) For some states and cities the number of total bicyclists captured in the ACS is very small. Additionally, disaggregating these estimates into male and female categories might lead to unexpected and unreliable results. For example, some cities show particularly low or high shares of women commuting by bicycle.

Legend:
 = High value
 = Low value

Walking to Work in States

State	% of commuters who walk to work	% men	% women	Percent of workers who walk to work by annual income (total = 100%)			
				<\$15,000	\$15,000-\$34,999	\$35,000-\$64,999	\$65,000+
Alabama	1.3%	60%	40%	51%	30%	13%	6%
Alaska	8.0%	63%	37%	34%	32%	22%	12%
Arizona	2.2%	58%	42%	44%	35%	14%	6%
Arkansas	1.8%	61%	39%	52%	28%	13%	6%
California	2.8%	53%	47%	43%	31%	15%	11%
Colorado	3.0%	56%	44%	48%	27%	16%	10%
Connecticut	2.9%	54%	46%	50%	24%	15%	10%
Delaware	2.4%	54%	46%	54%	22%	15%	9%
Florida	1.6%	56%	44%	45%	34%	14%	7%
Georgia	1.6%	59%	41%	46%	32%	13%	9%
Hawaii	4.7%	56%	44%	34%	40%	18%	8%
Idaho	3.1%	59%	41%	47%	32%	14%	7%
Illinois	3.2%	53%	47%	46%	26%	15%	12%
Indiana	2.2%	54%	46%	58%	26%	11%	5%
Iowa	4.0%	54%	46%	57%	24%	14%	5%
Kansas	2.7%	56%	44%	52%	29%	14%	5%
Kentucky	2.3%	57%	43%	55%	30%	10%	5%
Louisiana	2.0%	55%	45%	52%	31%	12%	5%
Maine	4.1%	55%	45%	48%	30%	16%	7%
Maryland	2.5%	51%	49%	48%	28%	14%	11%
Massachusetts	4.6%	48%	52%	42%	25%	19%	15%
Michigan	2.3%	53%	47%	58%	24%	12%	6%
Minnesota	3.0%	53%	47%	47%	26%	18%	9%
Mississippi	1.8%	62%	38%	54%	29%	11%	6%
Missouri	2.0%	57%	43%	53%	28%	13%	6%
Montana	5.2%	55%	45%	45%	30%	18%	7%
Nebraska	3.2%	56%	44%	49%	31%	15%	4%
Nevada	2.3%	53%	47%	40%	41%	12%	7%
New Hampshire	3.2%	52%	48%	50%	28%	14%	8%
New Jersey	3.3%	54%	46%	40%	33%	17%	10%
New Mexico	2.4%	57%	43%	47%	31%	15%	7%
New York	6.4%	50%	50%	35%	27%	19%	19%
North Carolina	1.9%	63%	37%	49%	32%	12%	7%
North Dakota	3.9%	57%	43%	46%	30%	16%	8%
Ohio	2.3%	53%	47%	57%	26%	12%	5%
Oklahoma	2.0%	61%	39%	54%	29%	12%	5%
Oregon	3.9%	52%	48%	46%	30%	15%	9%
Pennsylvania	4.0%	51%	49%	49%	28%	15%	8%
Rhode Island	3.1%	51%	49%	54%	21%	15%	10%
South Carolina	1.9%	58%	42%	55%	28%	10%	6%
South Dakota	4.7%	56%	44%	43%	34%	16%	6%
Tennessee	1.4%	61%	39%	55%	26%	13%	7%
Texas	1.8%	56%	44%	51%	31%	12%	6%
Utah	2.9%	52%	48%	62%	22%	12%	4%
Vermont	6.2%	50%	50%	50%	29%	14%	7%
Virginia	2.3%	58%	42%	45%	28%	14%	12%
Washington	3.5%	55%	45%	38%	33%	19%	10%
West Virginia	2.9%	57%	43%	57%	27%	12%	4%
Wisconsin	3.4%	53%	47%	53%	26%	14%	6%
Wyoming	4.0%	49%	51%	38%	35%	19%	9%
Mean/Average (1)	2.9%	54%	46%	46%	29%	15%	10%
Median	2.9%	55%	45%	49%	29%	14%	7%
High	8.0%	63%	52%	62%	41%	22%	19%
Low	1.3%	48%	37%	34%	21%	10%	4%

Legend:
 = High value
 = Low value

Source: ACS 2007-2009 Notes: (1) All averages are weighted.

Walking to Work in Cities

Cities (1)	% of commuters who walk to work	% men	% women	Percent of workers who walk to work by annual income (total = 100%)			
				<\$15,000	\$15,000-\$34,999	\$35,000-\$64,999	\$65,000+
Albuquerque	2.1%	49%	51%	53%	27%	16%	4%
Arlington, TX	1.9%	58%	42%	55%	32%	10%	3%
Atlanta	4.2%	62%	38%	40%	30%	15%	15%
Austin	2.0%	60%	40%	43%	32%	15%	10%
Baltimore	6.7%	46%	54%	39%	33%	18%	10%
Boston	13.9%	48%	52%	29%	22%	25%	25%
Charlotte	2.0%	61%	39%	39%	30%	13%	18%
Chicago	5.8%	48%	52%	35%	25%	20%	20%
Cleveland	4.4%	47%	53%	44%	33%	15%	8%
Colorado Springs	2.5%	57%	43%	53%	26%	13%	8%
Columbus	2.6%	60%	40%	60%	28%	9%	3%
Dallas	1.8%	48%	52%	42%	34%	16%	8%
Denver	4.1%	55%	45%	32%	28%	25%	15%
Detroit	3.3%	59%	41%	52%	33%	11%	4%
El Paso	2.2%	53%	47%	58%	32%	8%	1%
Fort Worth	1.2%	60%	40%	53%	30%	14%	2%
Fresno	2.0%	55%	45%	57%	32%	9%	3%
Honolulu	8.0%	53%	47%	28%	41%	20%	10%
Houston	2.2%	52%	48%	46%	33%	13%	8%
Indianapolis	2.1%	55%	45%	50%	31%	14%	5%
Jacksonville	1.5%	50%	50%	44%	34%	18%	5%
Kansas City, MO	2.1%	59%	41%	39%	38%	14%	9%
Las Vegas	2.0%	54%	46%	43%	43%	5%	10%
Long Beach	3.0%	55%	45%	52%	32%	9%	8%
Los Angeles	3.5%	52%	48%	48%	30%	14%	8%
Louisville	2.3%	47%	53%	52%	32%	9%	6%
Memphis	2.0%	59%	41%	62%	25%	11%	3%
Mesa	1.9%	70%	30%	34%	47%	16%	3%
Miami	3.8%	57%	43%	47%	32%	11%	10%
Milwaukee	4.6%	50%	50%	50%	28%	14%	7%
Minneapolis	6.4%	58%	42%	37%	29%	22%	12%
Nashville	1.7%	54%	46%	43%	21%	24%	12%
New Orleans	5.8%	60%	40%	41%	38%	14%	7%
New York	10.3%	49%	51%	26%	27%	22%	25%
Oakland	4.4%	53%	47%	38%	27%	23%	12%
Oklahoma City	1.4%	64%	36%	43%	37%	13%	7%
Omaha	2.7%	59%	41%	50%	33%	11%	6%
Philadelphia	8.4%	45%	55%	35%	28%	22%	15%
Phoenix	1.8%	58%	42%	38%	38%	16%	8%
Portland, OR	5.2%	53%	47%	43%	25%	20%	13%
Raleigh	2.6%	62%	38%	59%	31%	6%	5%
Sacramento	3.3%	51%	49%	28%	32%	21%	19%
San Antonio	2.0%	53%	47%	57%	28%	11%	4%
San Diego	2.9%	53%	47%	45%	30%	15%	10%
San Francisco	10.0%	54%	46%	23%	25%	25%	26%
San Jose	1.9%	54%	46%	34%	34%	19%	13%
Seattle	8.6%	53%	47%	28%	30%	25%	17%
Tucson	3.6%	54%	46%	51%	33%	13%	2%
Tulsa	2.2%	59%	41%	57%	30%	7%	6%
Virginia Beach	2.1%	58%	42%	56%	32%	7%	5%
Washington, DC	11.4%	50%	50%	20%	19%	25%	36%
Mean/ Average (2)	4.9%	51%	49%	35%	28%	19%	18%
Median	2.7%	54%	46%	43%	31%	14%	8%
High	13.9%	70%	55%	62%	47%	25%	36%
Low	1.2%	45%	30%	20%	19%	5%	1%

Legend:
 = High value
 = Low value

Source: ACS 2007-2009 Notes: (1) For some cities the number of total bicyclists captured in the ACS is very small. Additionally, disaggregating these estimates into male and female categories might lead to unexpected and unreliable results. For example, some cities show particularly low or high shares of women commuting by bicycle. (2) All averages are weighted.



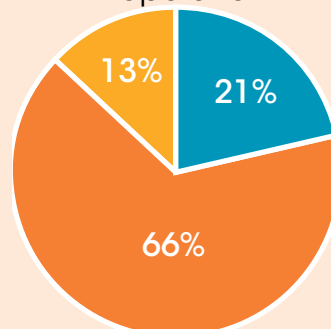
work comprising 7.1% of the working population and 9.9% of pedestrian commuters.

Age of Bicyclists and Pedestrians

It is no surprise that youth, who are not of legal driving age, make up a disproportionate amount of bicycling trips. National estimates from NHTS indicate that youth under age 16 make up 39% of bicycling trips, despite accounting for just 21% of the population. This age group accounts for 17% of walking trips. Adults over age 65 account for 13% of the population and make up 10% of all walking trips and 6% of all bicycling trips. The rest of people age 16-65 make up 66% of the population and account for 73% of all walking trips and 54% of trips by bicycle.

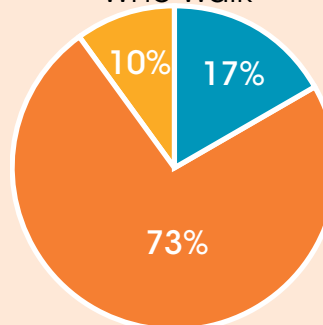
A Look at Age

Age of the U.S. Population



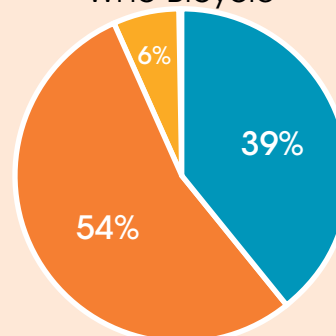
Source: ACS 2009

Age of People Who Walk



Source: NHTS 2009

Age of People Who Bicycle



Source: NHTS 2009

Note: (1) Numbers round down and so appear to equal 99%.

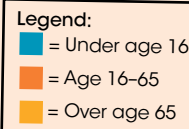




Photo by Eric Gilliland

3: SAFETY

While news headlines are filled with deaths of war and cancer victims, the public rarely hears reports on the more than 32,000 people who die each year in traffic crashes. In 2010, according to the National Highway Traffic Safety Administration (NHTSA), 32,788 people died on U.S. roadways. The death toll of pedestrians alone is equivalent to a jumbo jet full of passengers crashing roughly every month (Ernst 2011). Even though bicycle and pedestrian fatalities have been decreasing, bicyclists and pedestrians are still disproportionately at risk.

Data for this chapter came largely from the NHTSA's Fatality Analysis

Reporting System (FARS). FARS collects data from police reports of traffic accidents and is the authoritative national source for traffic fatalities in the United States. Data on bicycle and pedestrian injuries came from CDC's Web-based Injury Statistics Query and Reporting System (WISQARS).

FARS data indicate that bicyclists and pedestrians account for 13.5% of all traffic fatalities, despite the fact that they make up roughly 11.5% of all trips (according to NHTS estimates). In the 51 largest U.S. cities bicycling and walking account for 13.8% of all trips, yet bicyclists and pedestrians represent 30.0% of all traffic fatalities.

Victim Demographics

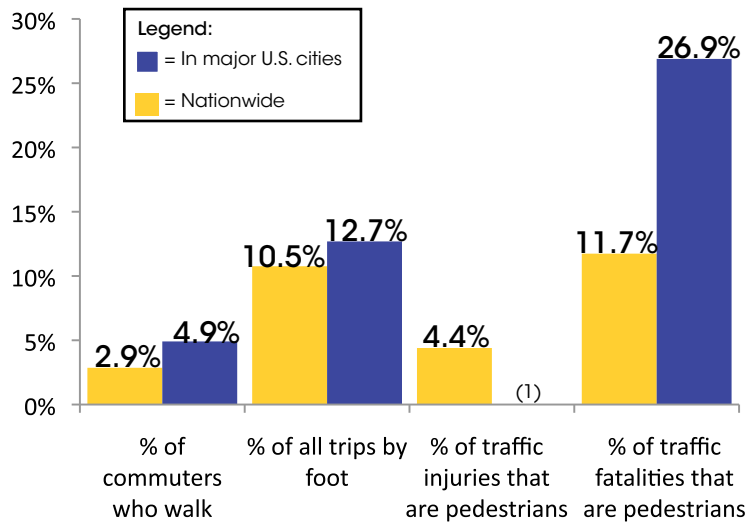
According to FARS, between 2007-2009, nearly one-quarter of all bicycle fatality victims are youth (under age 16) and seniors (over age 65). These age groups account for 26% of pedestrian fatalities. Seniors (over age 65) are at a disproportionate risk, accounting for just 10% of all walking trips and roughly 19% of pedestrian fatalities. Adults over age 65 make up 6% of all bicycling trips and account for roughly 10% of bicyclist fatalities.

In some areas the risk facing seniors is even greater. In Honolulu, where 46% of all traffic fatalities are pedestrians, 69% of victims are over age 65. Similarly in San Francisco, where 49% of all traffic fatalities are pedestrians, 50% of these are seniors. While cities do vary in their demographic composition, these rates of senior fatalities are still disproportionately higher than the percent of trips they represent.

What's the Risk?

To understand bicycle and pedestrian safety in a city or state, it is not enough to

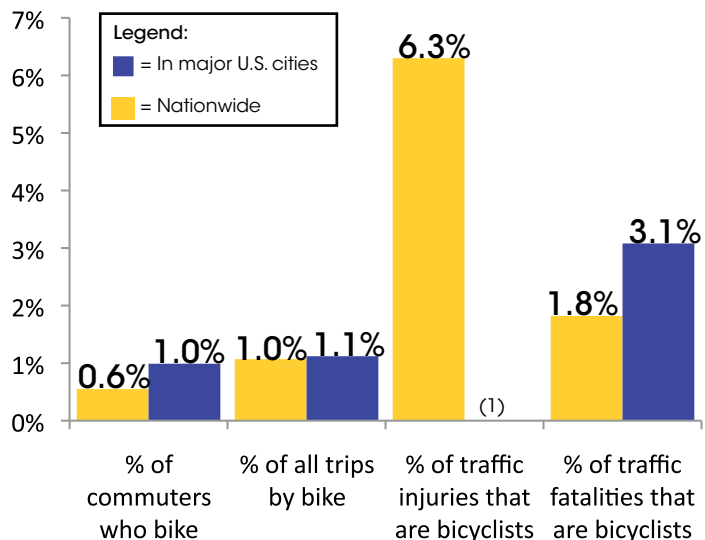
Overview of Walking and Pedestrian Safety Nationwide and in Largest U.S. Cities



Sources: FARS 2009, NHTS 2009, ACS 2009, WISQARS 2009

Note: (1) City-level data for pedestrian injuries is unavailable.

Overview of Bicycling and Bicycle Safety Nationwide and in Largest U.S. Cities



Sources: FARS 2009, NHTS 2009, ACS 2009, WISQARS 2009

Note: (1) City-level data for bicycle injuries is unavailable.